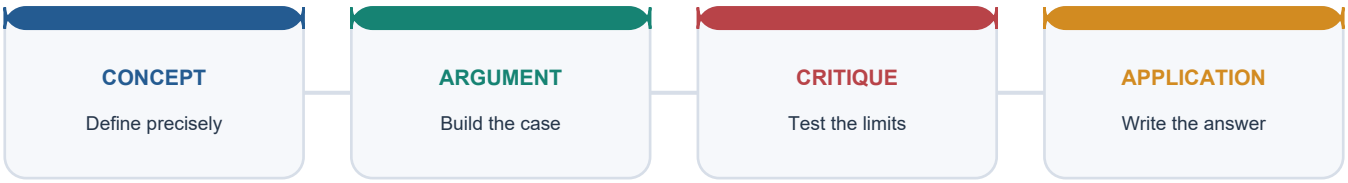
COMPLETE LEARNING SESSION

Concepts, Risk, Resilience and Sendai

GS-III Disaster Management | Concepts, risk, resilience, Sendai and answer writing

LEARNING PATH



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Concepts, Risk, Resilience and Sendai - Complete Learning Session

BASIC LEARNING SESSION

LEARNING SCOPE

Boundary	What stays distinct in this session
Risk vocabulary	Hazard, exposure, vulnerability, capacity, risk and disaster impact are never used as synonyms.
Cycle vocabulary	Prevention, mitigation, preparedness, relief, response, rehabilitation, reconstruction, recovery and Build Back Better are not conflated.
Treatment vocabulary	Risk acceptance, avoidance, reduction and transfer remain four separate strategies.
Plan vocabulary	The Disaster Management Act, 2005, the National Policy on Disaster Management, 2009 and the National Disaster Management Plan (2016, revised 2019) are separate instruments.
Sendai vocabulary	The four priorities for action are kept distinct from the seven lettered global targets, and no current-progress claim is made without a dated source.
Hazard boundary	Individual hazards are named only as illustrations; their operational management belongs to the hazard-specific topics.

This session teaches the conceptual grammar that every later disaster-management answer reuses. Each numbered session opens with its own visual, then defines the concept, explains it, demonstrates one examiner-grade paragraph, and closes with a trap and a recall line. The compact twelve-panel ASCII master flow at the end of the Consolidated Register Notes is the at-a-glance revision visual for the whole topic.

SESSION 1 - FOUNDATION - HAZARD, EXPOSURE, VULNERABILITY, CAPACITY AND THE DISASTER THRESHOLD

```
HAZARD          dangerous condition, process or event
   +
EXPOSURE        people, livelihoods, assets sitting in its path
   +
VULNERABILITY   conditions that make that exposure susceptible to harm
   /
CAPACITY        strengths, resources and measures that contain harm
   =
DISASTER RISK
   |
   |  if losses exceed the community's own ability to cope
   v
DISASTER = serious disruption of a community's functioning,
           causing human, material, economic and environmental loss
```

The rail shows the threshold: a hazard only becomes a disaster when exposure and vulnerability outrun capacity.

DEFINITION / WHAT THIS IS CALLED

Plain definition: Disaster risk has four building blocks - a dangerous event, who or what is in its path, why that exposure turns into harm, and what limits the harm.

Technical definition: A disaster is "a serious disruption to the functioning of a community, which causes human, material, economic and environmental losses beyond a community's ability to cope," arising from the combination of hazards, conditions of vulnerability and insufficient capacity or measures to reduce the potential negative consequences of risk.

ANSWER-GRABBING OPENING - WRITE/ADAPT IN THE EXAM

> A disaster is not an event but a threshold: the point at which a hazard meets exposed and vulnerable people whose capacity to cope has already been exhausted.

MUST-WRITE KEYWORDS

hazard | **exposure** | **vulnerability** | **capacity** | **coping ability** | **serious disruption**

CORE EXPLANATION

A hazard is a dangerous condition or event that threatens, or has the potential for causing, injury to life or damage to property or the environment. By itself it is only a potential. Exposure identifies the people, livelihoods, infrastructure and other assets actually located where the hazard can reach them. Vulnerability is the set of conditions - physical, social, economic, environmental and attitudinal - that increase susceptibility to the hazard's impact. Capacity is the combination of strengths, resources, institutions and measures available to manage and reduce the resulting risk.

The four terms are analytically separate and must stay separate in an answer. Their interaction produces disaster risk, and disaster risk crosses into disaster only when the resulting loss exceeds what the affected community can absorb on its own. That last clause - "beyond a community's ability to cope" - is the definitional hinge, because it explains why an identical hazard magnitude produces a non-event in one district and a catastrophe in the next.

WORKED EXAMPLE

A cyclone of a given wind speed crossing an uninhabited stretch of coast is a hazard with negligible exposure, so no disaster follows. The same wind speed crossing a densely settled coastal strip of poorly built housing, with no functioning shelter or evacuation plan, produces a disaster - the hazard did not change, but exposure rose, vulnerability was high and capacity was absent.

TECHNICAL DISTINCTION

Exposure answers *where and who*; vulnerability answers *why susceptible*. A high-income household may be highly exposed but only moderately vulnerable, while a poor household on the same street is both exposed and highly vulnerable. Collapsing the two removes the main policy lever, because exposure is reduced by location decisions and vulnerability by construction, income, health and inclusion measures.

NAMED-EVIDENCE PARAGRAPH DEMONSTRATION

Claim: Disaster impact is decided by the capacity threshold, not by hazard magnitude alone. **Named evidence:** The standard definition reproduced in India's disaster-management literature defines a disaster as serious disruption causing losses "beyond a community's ability to cope," and attributes it to the combination of hazards, conditions of vulnerability and insufficient capacity or risk-reduction measures. **Analysis:** Because "insufficient capacity" is written into the definition itself, the definition already identifies the intervention point - raising capacity and lowering vulnerability can prevent a disaster without altering the hazard at all. **Qualification:** The threshold is qualitative, not a measured cut-off; a specific event's classification still depends on assessment of the particular community's coping resources.

EXAMINER CAUTION

- Do not use "hazard" and "disaster" interchangeably; the marker is looking for the coping-capacity threshold.
- Do not describe exposure as if it were vulnerability - location and susceptibility are different questions with different remedies.

EXAM LINK

- **Prelims:** Keep hazard, exposure, vulnerability, capacity and disaster in five separate boxes when eliminating options.
- **Mains:** Define all four components before naming any measure; every measure targets one named component, never "disaster" in general.

MINI RECAP

- One chain: hazard meets exposure; vulnerability decides whether exposure becomes harm; capacity decides how much harm is contained.
- The definitional hinge is loss "beyond a community's ability to cope."

SESSION 2 - FOUNDATION - HAZARD TAXONOMY: NATURAL, SOCIO-NATURAL AND ANTHROPOGENIC FAMILIES

```
HAZARDS
  |
  +-- NATURAL
  |     +-- GEOPHYSICAL       earthquake, tsunami, volcanic activity, landslide
  |     +-- HYDROLOGICAL      flood, flash flood, debris flow
  |     +-- METEOROLOGICAL    cyclone, storm, extreme temperature event
  |     +-- CLIMATOLOGICAL    drought, desertification, forest fire
  |     +-- BIOLOGICAL        epidemic, pest infestation, animal attack
  |
  +-- SOCIO-NATURAL           natural process amplified by human action
  |                           (slope failure after hill cutting, urban flooding
  |                            after drainage encroachment)
  |
  +-- ANTHROPOGENIC           pollution, deforestation, chemical spillage,
                              industrial and transport accidents
```

The tree shows the classification by source that routes every later hazard chapter.

DEFINITION / WHAT THIS IS CALLED

Plain definition: Hazards are grouped by what produces them - nature, human action, or nature amplified by human action.

Technical definition: Classification by source separates natural hazards (geophysical, hydrological, climatological, meteorological and biological families) from anthropogenic hazards, with socio-natural hazards occupying the intermediate category where a natural process is created or intensified by human decisions.

ANSWER-GRABBING OPENING - WRITE/ADAPT IN THE EXAM

> Classification is not a decorative list: naming the hazard family fixes which science, which lead agency and which mitigation instrument the answer must then discuss.

MUST-WRITE KEYWORDS

geophysical | **hydrological** | **meteorological** | **climatological** | **biological** | **anthropogenic**

CORE EXPLANATION

The five natural families are not interchangeable. Geophysical hazards originate in earth processes, but their predictability differs: earthquake timing remains unavailable, while volcanoes and some landslides can be monitored for precursor conditions. Exposure control and construction or slope safety therefore remain central. Hydrological and meteorological hazards may be forecast at useful lead times, so early warning and evacuation are important. Climatological hazards such as drought and desertification develop over seasons, while biological hazards follow transmission dynamics and demand surveillance and public-health capacity.

Anthropogenic hazards - pollution, deforestation, chemical spillage and industrial accidents - arise directly from human activity and are governed primarily through regulation, siting and liability. The socio-natural category is the analytically important middle: the trigger is natural, but its severity is manufactured, as when hill-cutting destabilises a slope or built-over drainage converts ordinary rainfall into urban flooding.

WORKED EXAMPLE

Two settlements receive the same rainfall. In the first, natural drainage is intact and the event is a hydrological hazard with minor consequences. In the second, the drainage channel has been built over, so the same rainfall produces a socio-natural flood whose severity is attributable to a land-use decision rather than to the rainfall.

TECHNICAL DISTINCTION

A hazard is classified by its originating process, not by its severity or by who suffers. An epidemic is biological even when its spread is worsened by crowding; a chemical leak remains anthropogenic even when an earthquake triggers it, although the combined event is then treated as a cascading hazard chain.

NAMED-EVIDENCE PARAGRAPH DEMONSTRATION

Claim: The source classification is a policy routing device rather than a taxonomy exercise. **Named evidence:** The standard Indian disaster-management classification lists natural hazards under the geophysical, hydrological, climatological, meteorological and biological families and separately lists anthropogenic hazards including pollution, deforestation and chemical spillage. **Analysis:** Because each family has a different predictability profile, naming the family immediately commits the answer to the correct dominant instrument - construction safety for geophysical, warning and evacuation for hydro-meteorological, anticipatory relief for climatological, surveillance for biological and regulation for anthropogenic. **Qualification:** Many real events are cascading or compound, so classification identifies the primary driver and must not be used to deny secondary hazard chains.

EXAMINER CAUTION

- Do not treat "natural disaster" as a self-explanatory phrase; the loss component is rarely natural.
- Do not place drought among meteorological hazards when the question is testing the climatological, slow-developing family.

EXAM LINK

- **Prelims:** Family membership questions are commonly set on landslide (geophysical), drought (climatological) and epidemic (biological).
- **Mains:** Use the socio-natural category to argue that development decisions manufacture hazard severity, which sets up the risk-creation argument later in the paper.

MINI RECAP

- Five natural families, one anthropogenic set, and a socio-natural middle where human action amplifies a natural process.
- Family choice determines the dominant mitigation instrument.

SESSION 3 - FOUNDATION - ONSET SPEED: RAPID-ONSET EVENTS AND CREEPING EMERGENCIES

```
TIME AXIS ------------------------------------------------------------->

RAPID ONSET      |####|
                 earthquake, cyclone, flood, tsunami
                 minutes to days | acute, visible, mediatised
                 response system: warning, evacuation, search and rescue

SLOW ONSET       |##############################################|
"CREEPING        climate change, desertification, soil degradation, drought
 EMERGENCIES"    months to years | cumulative, low-visibility
                 loss mechanism: progressive loss of sustenance
                 response system: anticipatory planning, livelihood support
```

The strip contrasts an acute shock with a cumulative erosion of the ability to derive sustenance.

DEFINITION / WHAT THIS IS CALLED

Plain definition: Some disasters strike in minutes; others build up over years without a single visible moment of impact.

Technical definition: Rapid-onset disasters such as earthquake, cyclone, flood and tsunami produce sudden acute impact, while slow-onset disasters - termed "creeping emergencies" and including climate change, desertification, soil degradation and drought - unfold over months or years and must find adequate reflection in disaster preparedness.

ANSWER-GRABBING OPENING - WRITE/ADAPT IN THE EXAM

> Creeping emergencies are the category institutional machinery is least designed for, because they never generate the single dramatic moment around which response systems are built.

MUST-WRITE KEYWORDS

rapid onset | **slow onset** | **creeping emergencies** | **desertification** | **loss of sustenance** | **preparedness reflection**

CORE EXPLANATION

The onset classification matters because it changes the shape of the response. A rapid-onset event has a definable impact moment, so the system that matters is the warning-to-evacuation-to-rescue chain, and success is measured in hours. A slow-onset process has no impact moment; societies lose their ability to derive sustenance over a period of time, so the system that matters is anticipatory planning, livelihood protection, water and land management, and social protection, and success is measured across seasons.

The source material is explicit that slow-onset disasters must find adequate reflection in disaster preparedness, precisely because their gradual character makes them easy to defer. An answer that treats preparedness as synonymous with evacuation drills therefore misses half the category.

WORKED EXAMPLE

A cyclone landfall triggers a defined sequence - bulletin, warning, evacuation, shelter, relief - completed within a week. A multi-year drought in the same State triggers no such sequence; instead, the household sells assets, withdraws a child from school and migrates, and the loss is recorded only as distress migration rather than as disaster loss.

TECHNICAL DISTINCTION

Onset speed is not the same as duration of consequences. A rapid-onset earthquake can have decade-long recovery consequences, yet remains rapid-onset because the hazard impact itself is instantaneous. Conversely, a cyclone that intensifies over two days is still rapid-onset relative to a creeping emergency measured in years.

NAMED-EVIDENCE PARAGRAPH DEMONSTRATION

Claim: Slow-onset disasters are structurally under-served by disaster institutions built around acute events. **Named evidence:** The standard treatment of onset speed classifies climate change, desertification, soil degradation and drought as "creeping emergencies", states that societies lose their ability to derive sustenance over a period of time, and requires that such disasters find adequate reflection in preparedness planning. **Analysis:** Because these losses accumulate without a declarable impact moment, they do not trigger the warning, rescue and relief machinery that dominates institutional design, and the resulting policy attention gap is itself an examinable argument. **Qualification:** This is an argument about institutional design emphasis, not a claim that India has no drought or desertification programmes; those exist under separate developmental mandates.

EXAMINER CAUTION

- Do not classify flash floods or rapidly intensifying cyclones as creeping emergencies simply because they develop over some hours.
- Do not equate slow onset with low severity; cumulative losses can exceed those of a single acute event.

EXAM LINK

- **Prelims:** The tested pairing is rapid onset (earthquake, cyclone, flood, tsunami) against slow onset (drought, desertification, soil degradation, climate change).
- **Mains:** Use the slow-onset gap as a ready counterpoint in any "critically examine preparedness" demand.

MINI RECAP

- Rapid onset has an impact moment; creeping emergencies erode sustenance without one.
- Preparedness must cover both, which is exactly where institutional design is weakest.

SESSION 4 - FOUNDATION - THE FIVE-FOLD VULNERABILITY TYPOLOGY

DIMENSION	WHAT MAKES THE COMMUNITY SUSCEPTIBLE
ECONOMIC	narrow income diversity, weak access to the means of production, no savings or insurance
PHYSICAL	proximity to the hazard, type and quality of construction, unsafe siting
SOCIAL	the poor, pregnant and lactating women, persons with disability, children and the elderly
ENVIRONMENTAL	sensitivity and degradation of the surrounding ecosystem, loss of natural buffers
ATTITUDINAL	a community's negative disposition towards change, producing dependence on external support

The five-row matrix is the enumerable typology a vulnerability question is marked against.

DEFINITION / WHAT THIS IS CALLED

Plain definition: Vulnerability is the set of conditions that make the same event hurt some people far more than others.

Technical definition: Vulnerability comprises "conditions determined by physical, social, economic, and environmental factors or processes, which increase the susceptibility of a community to the impact of hazards", taking economic, physical, social, environmental and attitudinal forms.

ANSWER-GRABBING OPENING - WRITE/ADAPT IN THE EXAM

> Vulnerability, unlike hazard frequency, is produced by decisions - which is precisely why it is the component of disaster risk that policy can actually move.

MUST-WRITE KEYWORDS

economic vulnerability | **physical vulnerability** | **social vulnerability** | **environmental vulnerability** | **attitudinal vulnerability** | **susceptibility**

CORE EXPLANATION

Economic vulnerability turns on income diversity and access to the means of production: a household with a single rain-dependent livelihood has no fallback when that livelihood fails. Physical vulnerability turns on proximity and construction type: the same ground motion collapses unreinforced masonry and leaves an engineered frame standing. Social vulnerability is defined by who cannot act on a warning or absorb a loss - the poor, pregnant and lactating women, persons with disability, children and the elderly. Environmental vulnerability captures ecosystem sensitivity, where degraded mangroves, wetlands or slope cover no longer buffer the hazard.

Attitudinal vulnerability is the least-used and most distinguishing of the five: a community's negative disposition towards change, which produces dependence on external support rather than self-protective action. Naming it signals that the candidate has read the full typology rather than the usual four.

WORKED EXAMPLE

In one coastal ward, a landless daily-wage household (economic) lives in a thatch dwelling on the seaward side (physical), headed by an elderly widow (social), behind a mangrove belt cleared for aquaculture (environmental), in a settlement that has repeatedly declined shelter relocation on the belief that relief will arrive anyway (attitudinal). The five dimensions coincide, so the harm is compounding rather than merely additive.

TECHNICAL DISTINCTION

Vulnerability is a condition of the exposed unit, not a property of the hazard and not the same as poverty. Poverty strongly drives economic and social vulnerability, but a wealthy settlement built on an unstable slope is physically vulnerable, and a well-resourced community that rejects risk information is attitudinally vulnerable.

NAMED-EVIDENCE PARAGRAPH DEMONSTRATION

Claim: A complete vulnerability answer must run to five dimensions, not the conventional four. **Named evidence:** The five-fold typology names economic vulnerability in terms of income diversity and access to means of production, physical vulnerability in terms of proximity and construction type, social vulnerability in terms of the poor, pregnant and lactating women, the disabled, children and the elderly, environmental vulnerability in terms of ecosystem sensitivity, and attitudinal vulnerability in terms of a community's response to change. **Analysis:** The attitudinal dimension converts vulnerability from a purely material condition into a behavioural one, which is what justifies awareness, participation and community-based programmes as genuine risk-reduction rather than as soft add-ons. **Qualification:** Attitudinal vulnerability must be argued as a product of past experience and exclusion, not used to blame affected communities for their own losses.

EXAMINER CAUTION

- Do not stop at four dimensions; the attitudinal dimension is the mark-earning addition.
- Do not treat vulnerability as fixed; it is policy-responsive alongside exposure and capacity.

EXAM LINK

- **Prelims:** The commonly set question asks which further dimension is added beyond physical, social, economic and environmental.
- **Mains:** The 2019 GS-III question is answered directly from this typology, with one concrete example per dimension.

MINI RECAP

- Five dimensions: economic, physical, social, environmental, attitudinal.
- Coinciding dimensions compound harm; vulnerability can be reduced even when hazard frequency cannot, while exposure and capacity remain additional policy levers.

SESSION 5 - CORE - RISK AS EXPECTED LOSS AND THE WORKING RELATION

```
DEFINITION
  RISK = a measure of the EXPECTED LOSSES due to a hazard event
         occurring in a GIVEN AREA over a SPECIFIC TIME PERIOD

WORKING RELATION
  RISK  =  PROBABILITY OF HAZARD  x  DEGREE OF VULNERABILITY

USE IT TO                              DO NOT USE IT TO
  order the components of a problem      produce an exact numerical
  identify which lever can be moved       forecast of loss
  compare two settlements                 claim a precise prediction
  justify capacity investment             of the next event
```

The panel separates the definition, the working relation and the limit of that relation.

DEFINITION / WHAT THIS IS CALLED

Plain definition: Risk is the loss you should expect in a stated place over a stated period, not the event itself.

Technical definition: Risk is "a measure of the expected losses due to a hazard event occurring in a given area over a specific time period", commonly organised through the working relation Risk = Probability of Hazard x Degree of Vulnerability.

ANSWER-GRABBING OPENING - WRITE/ADAPT IN THE EXAM

> The risk relation earns marks as a disciplined heuristic that locates the policy lever; it loses them the moment it is presented as an arithmetic that predicts the next event.

MUST-WRITE KEYWORDS

expected losses | **given area** | **specific time period** | **probability of hazard** | **degree of vulnerability** | **working relation**

CORE EXPLANATION

Three elements of the definition are examinable and routinely dropped. First, risk is about *expected* loss, which makes it a forward-looking estimate rather than a record of past damage. Second, it is bounded by a *given area*, so a national risk statement is meaningless without a spatial unit. Third, it is bounded by a *specific time period*, so a risk figure without a return period or planning horizon is incomplete.

The working relation multiplies hazard probability by degree of vulnerability. Its analytical value is that it shows a fall in either factor lowers risk, and that where hazard probability cannot be altered - as with earthquakes - the whole reduction must come from the vulnerability side, moderated by the capacity that measures and institutions supply. Its limit is equally important: it is a relation for organising reasoning, not a validated predictive formula applicable to every event.

WORKED EXAMPLE

Two towns face the same one-in-fifty-year flood probability. Town A has raised plinths, an enforced building code and a functioning warning chain; Town B has none. Hazard probability is identical, so the entire risk difference is attributable to the vulnerability and capacity terms - which is exactly the sentence an examiner is looking for.

TECHNICAL DISTINCTION

Risk is not the same as hazard, and residual risk is not the same as zero risk. After every feasible reduction measure, the risk that remains is residual risk, which must then be accepted, transferred or prepared for. Confusing "risk reduced" with "risk eliminated" is the most common conceptual slip in this area.

NAMED-EVIDENCE PARAGRAPH DEMONSTRATION

Claim: The risk relation should be deployed as an analytical discipline rather than as a formula. **Named evidence:** The accepted definition treats risk as a measure of expected losses in a given area over a specific time period and then offers Risk = Probability of Hazard x Degree of Vulnerability as a working relation rather than as a measurement standard. **Analysis:** Read together, the definition and the relation tell the candidate to state a spatial unit and a time horizon, then to identify which of the two factors is actually movable in the case under discussion, which converts a definition into an argument. **Qualification:** The relation omits exposure and capacity as explicit terms, so it should be presented as a simplified organising device supplemented by the four-component framework of Session 1.

EXAMINER CAUTION

- Do not state a risk claim without naming the area and the time period it refers to.
- Do not convert the working relation into a decimal figure and then treat that figure as a forecast.

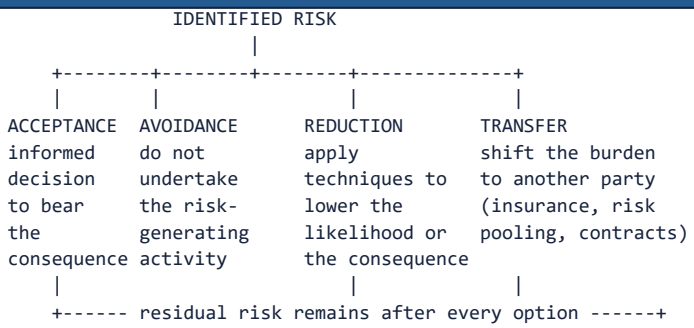
EXAM LINK

- **Prelims:** Options that describe the relation as a precise universal law, or as superseded by Sendai, are distractors.
- **Mains:** Open any risk question by defining risk with its area and time bounds, then use the relation to identify the reducible factor.

MINI RECAP

- Expected loss, in a given area, over a specific time period.
- Probability multiplied by vulnerability is a working relation; residual risk always remains.

SESSION 6 - CORE - THE FOUR WAYS OF DEALING WITH RISK



Four distinct strategies, only one of which actually lowers the physical likelihood or consequence.

DEFINITION / WHAT THIS IS CALLED

Plain definition: Once a risk is identified, there are only four honest things to do with it: live with it, step away from it, shrink it, or pay someone else to carry it.

Technical definition: Risk acceptance is an informed decision to accept the consequences; risk avoidance means not undertaking the risk-generating activity; risk reduction applies techniques to reduce the likelihood or the consequence of a loss; risk transfer shifts the burden of loss to another party, for example through insurance.

ANSWER-GRABBING OPENING - WRITE/ADAPT IN THE EXAM

> Risk reduction and risk transfer are not synonyms: one lowers the loss that will occur, the other decides who pays for it once it has.

MUST-WRITE KEYWORDS

risk acceptance | **risk avoidance** | **risk reduction** | **risk transfer** | **residual risk** | **informed decision**

CORE EXPLANATION

Acceptance is legitimate only when it is informed: a community that knowingly bears a small, infrequent loss because the cost of protection exceeds the expected damage has accepted risk; a community that is simply unaware has not. Avoidance is the strongest option where it is feasible, because it removes the exposure entirely, as when construction is prohibited in an active floodway. Reduction is the mainstream instrument of disaster management, covering everything from embankments and retrofitting to zoning and codes. Transfer neither prevents nor reduces the physical loss; it redistributes the financial consequence through insurance, pooled funds or contractual allocation.

The four options are complements rather than rivals. A sound risk-treatment package avoids what can be avoided, reduces what can be reduced, transfers a defined share of what remains, and consciously accepts the small residue.

WORKED EXAMPLE

For a coastal industrial estate, avoidance means not siting new units seaward of the regulated line; reduction means elevating plinths and anchoring storage tanks; transfer means catastrophe insurance for the remaining asset loss; acceptance means the operator knowingly carrying minor annual inundation losses that cost less to absorb than to prevent.

TECHNICAL DISTINCTION

Transfer is frequently mislabelled as reduction in written answers. Insurance leaves hazard probability and physical vulnerability untouched, so total societal loss is unchanged; what changes is the speed and fairness of the financial recovery. Only reduction and avoidance alter the loss that actually occurs.

NAMED-EVIDENCE PARAGRAPH DEMONSTRATION

Claim: The four-option framework, not a general call for "more mitigation", is what gives a risk-treatment answer analytical structure. **Named evidence:** Standard risk-treatment doctrine separates risk acceptance, risk avoidance, risk reduction and risk transfer as four distinct ways of dealing with risk, defining transfer specifically as shifting the burden of loss to another party such as an insurer. **Analysis:** Once the four are separated, an answer can diagnose the actual gap - typically that India's system is strong on post-event compensation and comparatively thin on ex-ante avoidance and transfer - instead of recommending unspecified mitigation. **Qualification:** The relative weight of each option is a design judgement that depends on the hazard, the asset and the financing capacity, so no single option should be presented as universally superior.

EXAMINER CAUTION

- Do not describe insurance as risk reduction; it is risk transfer.
- Do not treat acceptance as negligence; the source defines it as an informed decision.

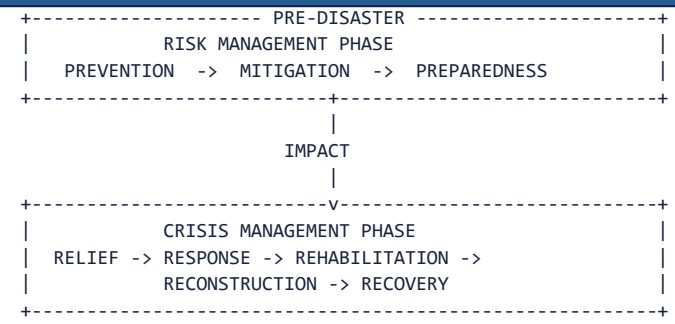
EXAM LINK

- **Prelims:** Scenario options usually test insurance (transfer) against relocation (avoidance) and retrofitting (reduction).
- **Mains:** Use the four options as a diagnostic grid when a question asks how disaster risk should be "managed" rather than merely reduced.

MINI RECAP

- Acceptance, avoidance, reduction, transfer - four distinct strategies.
- Only avoidance and reduction change the loss; transfer changes who bears it.

SESSION 7 - CORE - THE DISASTER MANAGEMENT CYCLE: TWO PHASES AND THE SHIFT FROM 3Rs TO 3Ps




```
APPROACH SHIFT
  OLD  3 Rs :  RESCUE       RELIEF        RESTORATION     (reactive)
  NEW  3 Ps :  PREVENTION   PREPAREDNESS  PROOFING        (anticipatory)
```

The cycle has two named phases, and the 3Rs-to-3Ps shift is the change of philosophy between them.

DEFINITION / WHAT THIS IS CALLED

Plain definition: The cycle splits into what is done before impact and what is done after it, and modern policy has moved the centre of gravity to the "before".

Technical definition: The disaster management cycle comprises a pre-disaster Risk Management Phase of prevention, mitigation and preparedness and a post-disaster Crisis Management Phase of relief, response, rehabilitation, reconstruction and recovery; the traditional reactive "3 Rs" of Rescue, Relief and Restoration have been replaced by the "3 Ps" of Prevention, Preparedness and Proofing.

ANSWER-GRABBING OPENING - WRITE/ADAPT IN THE EXAM

> The move from the 3 Rs to the 3 Ps is not a change of vocabulary but a change of where money, authority and attention sit in the cycle.

MUST-WRITE KEYWORDS

Risk Management Phase | **Crisis Management Phase** | **Rescue, Relief, Restoration** | **Prevention, Preparedness, Proofing** | **pre-disaster** | **post-disaster**

CORE EXPLANATION

The Risk Management Phase is where risk is actually altered. Prevention removes it, mitigation lessens it, and preparedness builds the ability to act when the residue arrives. The Crisis Management Phase begins at impact and runs from life-saving relief and response, through rehabilitation of services and livelihoods, to reconstruction of assets and full recovery.

The 3 Rs describe a system that waited for the event and then organised rescue, relief and restoration. Restoration is the revealing word, because it means putting the situation back as it was, which necessarily reinstates the vulnerability that produced the loss. The 3 Ps replace it with proofing - making people, assets and systems capable of withstanding the next event - and put prevention and preparedness ahead of it in sequence. The shift therefore reverses the direction of the cycle: instead of the response phase driving the system, the risk-management phase does.

WORKED EXAMPLE

After repeated flooding of a market town, a 3 Rs administration evacuates, distributes relief and rebuilds the same shops on the same plinth level. A 3 Ps administration prohibits rebuilding in the active channel, raises plinths and drainage capacity, pre-positions stocks and runs an annual drill, so the next event of similar magnitude produces a smaller loss.

TECHNICAL DISTINCTION

Relief is not recovery and restoration is not reconstruction. Relief meets immediate survival needs; rehabilitation restores services and livelihoods; reconstruction rebuilds physical assets; recovery is the process of restoring or improving functioning, and is the phase in which risk can either be reproduced or reduced.

NAMED-EVIDENCE PARAGRAPH DEMONSTRATION

Claim: The 3Rs-to-3Ps shift is the sharpest available evidence that India's disaster policy has moved from a relief-centric to a risk-centric model. **Named evidence:** The standard account explicitly structures the cycle into a pre-disaster Risk Management Phase and a post-disaster Crisis Management Phase, and records the replacement of the traditional 3 Rs - Rescue, Relief and Restoration - with the 3 Ps of Prevention, Preparedness and Proofing. **Analysis:** Because "restoration" is replaced by "proofing", the new formulation writes risk reduction into recovery itself, which is the same logic later expressed globally as Build Back Better. **Qualification:** The shift describes a declared approach; whether budget shares and institutional effort have moved correspondingly requires separate, dated evidence.

EXAMINER CAUTION

- Do not list the cycle phases without naming the two governing phases; the phase names are the scoring detail.
- Do not present the 3 Ps as an addition to the 3 Rs; the source frames it as a replacement of a reactive approach.

EXAM LINK

- **Prelims:** The tested facts are the composition of each phase and the exact expansion of the 3 Rs and 3 Ps.
- **Mains:** Use the phase split to locate the question precisely - a warning question sits in preparedness, a compensation question in recovery.

MINI RECAP

- Risk Management Phase before impact; Crisis Management Phase after it.
- Rescue, Relief, Restoration gives way to Prevention, Preparedness, Proofing.

SESSION 8 - CORE - PREVENTION, MITIGATION AND PREPAREDNESS: THREE SEPARATE PRE-IMPACT ACTIONS

ACTION	QUESTION IT ANSWERS	TYPICAL MEASURE
PREVENTION	Can the risk be avoided altogether?	prohibit building in an active floodway; relocate a settlement off a fault
MITIGATION	If not, can its impact be lessened?	STRUCTURAL: embankment, retrofit, cyclone shelter NON-STRUCTURAL: land-use zoning, building codes
PREPAREDNESS	When it still arrives, can we act in time?	contingency plans, stocks, early warning, drills, trained response teams

Three different questions, three different instruments, one temporal sequence.

DEFINITION / WHAT THIS IS CALLED

Plain definition: Prevention avoids the risk, mitigation shrinks the damage, preparedness gets you ready to act.

Technical definition: Prevention seeks the outright avoidance of adverse impacts where avoidance is feasible; mitigation lessens the adverse impacts of a hazardous event through structural and non-structural measures; preparedness develops the knowledge and capacities of governments, responders and communities to anticipate, respond to and recover from an impending event.

ANSWER-GRABBING OPENING - WRITE/ADAPT IN THE EXAM

> The three pre-impact actions answer three different questions, so an answer that uses them interchangeably has already conceded the structure mark.

MUST-WRITE KEYWORDS

prevention | mitigation | structural measures | non-structural measures | preparedness | contingency planning

CORE EXPLANATION

Prevention is the most powerful and the least frequently available option, because avoidance usually requires controlling location, and location is governed by land markets, livelihoods and existing settlement. Mitigation accepts that the hazard will occur and reduces what it can do. Its structural arm builds physical defences and strengthens assets; its non-structural arm uses regulation, zoning, codes, insurance conditions and awareness to change how and where building happens. Preparedness has the shortest horizon: plans, stockpiles, warning systems, drills and trained teams that convert a warning into a completed evacuation.

Their relationship is sequential rather than substitutable. Preparedness cannot compensate for absent mitigation, because a perfectly executed evacuation still leaves the assets to be destroyed; mitigation cannot compensate for absent preparedness, because a strengthened building does not itself move people out of a flood path.

WORKED EXAMPLE

For an unstable hill slope above a highway, prevention is realigning the road away from the slope; mitigation is slope stabilisation and retaining structures, backed by a zoning restriction on new construction; preparedness is a rainfall-threshold warning, a closure protocol and a pre-designated diversion route.

TECHNICAL DISTINCTION

Relocation is prevention, not mitigation, because it removes the exposure rather than reducing the damage to an exposed asset. Conversely, a cyclone shelter is mitigation in its construction quality and preparedness in its staffing, stocking and evacuation plan, which is why the boundary is analytically useful rather than rigid.

NAMED-EVIDENCE PARAGRAPH DEMONSTRATION

Claim: Structural and non-structural mitigation must both appear, because a structural-only answer describes engineering rather than disaster management. **Named evidence:** The Risk Management Phase places prevention, mitigation and preparedness before impact, and the mitigation literature it draws on divides measures into structural works such as embankments and retrofitting and non-structural instruments such as land-use zoning and building codes. **Analysis:** The non-structural arm is where risk creation is actually stopped, since a code that is enforced prevents new vulnerable stock from being added faster than old stock can be retrofitted. **Qualification:** Non-structural measures depend entirely on enforcement capacity, so their presence in a plan is not evidence of their effect on the ground.

EXAMINER CAUTION

- Do not classify relocation as mitigation; avoidance of exposure is prevention.
- Do not list only embankments and shelters; omitting non-structural measures truncates the mitigation answer.

EXAM LINK

- **Prelims:** Questions commonly ask which listed activity is preparedness rather than mitigation.
- **Mains:** Allocate one sentence to each of the three, with a named measure, before moving to the post-impact phase.

MINI RECAP

- Prevention avoids, mitigation lessens, preparedness readies.
- Mitigation always has a structural and a non-structural arm.

SESSION 9 - CORE - AFTER IMPACT: RELIEF, RESPONSE, REHABILITATION, RECONSTRUCTION AND BUILD BACK BETTER

```
IMPACT
  |
  v
RELIEF AND RESPONSE        hours to weeks
  search and rescue, medical care, food, water, shelter, restoring
  life-saving services
  |
  v
REHABILITATION             weeks to months
  restore basic services, schooling, health access and livelihoods
  |
  v
RECONSTRUCTION             months to years
  rebuild housing, infrastructure and productive assets
  |
  v
RECOVERY                   process of restoring or improving functioning
  |
  +--> RESTORE AS BEFORE   -> the same vulnerability returns
  |
  +--> BUILD BACK BETTER   -> safer siting, stronger codes, restored
                              ecosystems, protected livelihoods
```

The fork at the end of the ladder is where the next disaster is either created or prevented.

DEFINITION / WHAT THIS IS CALLED

Plain definition: After the event, the system moves from saving lives to restoring services to rebuilding assets, and at the last step it chooses whether to rebuild the old risk.

Technical definition: The Crisis Management Phase comprises relief and response, rehabilitation, reconstruction and recovery; Build Back Better is the principle that recovery and reconstruction should reduce future risk rather than reproduce pre-existing vulnerability.

ANSWER-GRABBING OPENING - WRITE/ADAPT IN THE EXAM

> Reconstruction is the only moment when finance, political attention and public consent for change are simultaneously available, which is why Build Back Better is a governance opportunity and not a slogan.

MUST-WRITE KEYWORDS

relief | **response** | **rehabilitation** | **reconstruction** | **recovery** | **Build Back Better**

CORE EXPLANATION

Relief and response are life-safety operations measured in hours and days: search and rescue, medical care, emergency shelter, water and restoration of critical services. Rehabilitation follows, restoring schooling, health access, markets and livelihoods so that households stop depending on external supply. Reconstruction rebuilds the damaged physical stock. Recovery is the process of restoring or improving livelihoods, health, assets, systems and activities of an affected community in ways that support sustainable development and reduce future risk. It is therefore the phase in which the risk profile for the next event is set.

Build Back Better attaches a condition to that final stage. Instead of restoring the damaged asset to its previous specification, reconstruction must relocate or redesign where siting was unsafe, raise the standard where the code was inadequate, restore the ecosystem buffers that were lost, and rebuild livelihoods on a less fragile base. Without that condition, an efficient reconstruction simply resets the system for the next event of the same magnitude.

WORKED EXAMPLE

A flood destroys a riverside colony. Reconstruction to the previous specification returns identical houses to the identical floodway within eighteen months. Reconstruction under Build Back Better shifts the colony behind the flood line, upgrades the drainage design standard, restores the wetland that was encroached, and funds livelihood replacement so that households do not return informally to the old site.

TECHNICAL DISTINCTION

Recovery is not a synonym for reconstruction. Physical assets can be fully reconstructed while livelihoods, schooling and health access remain unrecovered, which is why rehabilitation and reconstruction are separately named and why recovery outcomes require their own evidence.

NAMED-EVIDENCE PARAGRAPH DEMONSTRATION

Claim: Recovery decisions, not response speed, determine the risk carried into the next cycle. **Named evidence:** The standard cycle places relief, response, rehabilitation, reconstruction and recovery inside a single post-disaster Crisis Management Phase and, in replacing "Restoration" with "Proofing" in the 3Ps formulation, requires that the rebuilt state be more hazard-resistant than the state that failed. **Analysis:** This makes reconstruction a risk-reduction instrument in its own right, so a recovery answer should be evaluated by whether the rebuilt system is safer, not by how quickly normality was declared. **Qualification:** Build Back Better depends on financing, land availability and enforcement follow-through, so its appearance in a policy document is an input, not a demonstrated outcome.

EXAMINER CAUTION

- Do not present Build Back Better as merely faster or better-quality reconstruction; the defining element is reduced future risk.
- Do not treat relief expenditure as evidence of recovery.

EXAM LINK

- **Prelims:** The tested distinction is between rehabilitation (services and livelihoods) and reconstruction (physical assets).
- **Mains:** Close any cycle-based answer on the recovery fork, because it converts description into evaluation.

MINI RECAP

- Relief and response, then rehabilitation, then reconstruction, then recovery.
- The recovery fork decides whether the old vulnerability returns.

SESSION 10 - CORE - DISASTER RISK REDUCTION: DEFINITION AND FIELD OF ACTION

```
DRR DEFINITION
  reducing disaster risks through
    (1) reduced EXPOSURE to hazards
    (2) lessened VULNERABILITY of people and property
    (3) wise management of LAND and the ENVIRONMENT
    (4) improved PREPAREDNESS for adverse events

FIELD OF ACTION  (a sequence with feedback, not a menu)

  POLICY AND LEGAL FRAMEWORK
        v
  RISK ASSESSMENT ------------------------+
        v                                 |
  RISK AWARENESS                          | feedback:
        v                                 | new data revises
  PLAN IMPLEMENTATION                     | assessment and plans
        v                                 |
  EARLY WARNING SYSTEMS                   |
        v                                 |
  USE OF KNOWLEDGE ----------------------+
```

Skipping risk assessment early in the chain degrades every stage that follows it.

DEFINITION / WHAT THIS IS CALLED

Plain definition: Disaster risk reduction is the whole practice of making disasters smaller before they happen.

Technical definition: DRR is "the concept and practice of reducing disaster risks through reduced exposure to hazards, lessened vulnerability of people and property, wise management of land and the environment, and improved preparedness", operating through a policy framework, risk assessment, risk awareness, plan implementation, early warning systems and the use of knowledge.

ANSWER-GRABBING OPENING - WRITE/ADAPT IN THE EXAM

> Disaster risk reduction is a development practice before it is an emergency practice: three of its four defining routes run through land, property and environment, not through response.

MUST-WRITE KEYWORDS

reduced exposure | lessened vulnerability | wise land management | improved preparedness | risk assessment | early warning systems

CORE EXPLANATION

The four-part definition is worth reproducing in full because each part names a different ministry's business. Reduced exposure is a planning and settlement question. Lessened vulnerability of people and property is a housing, livelihood and social-protection question. Wise management of land and the environment is an ecological question, where wetlands, mangroves, forests and slopes function as risk-reducing infrastructure. Improved preparedness is the only part that belongs primarily to the disaster-management machinery itself.

The field of action is best read as an ordered sequence with feedback. A policy and legal framework creates the mandate; risk assessment supplies the information; risk awareness converts that information into public and institutional understanding; plan implementation turns it into works and rules; early warning systems handle the residual risk that remains; and the use of knowledge, including local and traditional knowledge, feeds experience back into the next assessment. Because the stages are dependent, a system that legislates and warns but never assesses risk properly will implement plans against the wrong problem.

WORKED EXAMPLE

A district with a disaster management act and an alert system but no hazard mapping issues warnings for the river while the recurring losses come from unmapped urban drainage failure. The missing stage is risk assessment, and no amount of warning capacity compensates for it.

TECHNICAL DISTINCTION

DRR is the objective and policy practice; disaster management is the organisational system that delivers it. DRR also has a stated three-part scope - preventing new risk, reducing existing risk and managing residual risk - which maps precisely onto the prevention, mitigation and preparedness-plus-transfer instruments already taught.

NAMED-EVIDENCE PARAGRAPH DEMONSTRATION

Claim: The DRR field of action is a dependent sequence, so identifying the broken stage is more valuable than listing all six. **Named evidence:** The accepted statement of disaster risk reduction sets out the field of action as policy and legal framework, risk assessment, risk awareness, plan implementation, early warning systems and use of knowledge, and defines DRR itself through reduced exposure, lessened vulnerability, wise land and environmental management and improved preparedness. **Analysis:** Because the later stages consume the outputs of the earlier ones, an answer that diagnoses which stage has failed in the case being discussed demonstrates a command of the framework that a recitation cannot. **Qualification:** The sequence is an analytical ordering rather than a statutory procedure, and in practice the stages run concurrently and iteratively.

EXAMINER CAUTION

- Do not reduce DRR to preparedness and early warning; exposure, vulnerability and environmental management carry equal weight in the definition.
- Do not present the field of action as an unordered list of independent options.

EXAM LINK

- **Prelims:** The sequence of the field of action and the four routes in the definition are both directly testable.
- **Mains:** Use the definition's four routes as ready-made paragraph headings for any "how can disaster risk be reduced" demand.

MINI RECAP

- Four routes: exposure, vulnerability, land and environment, preparedness.
- Six stages in sequence with feedback, beginning with the legal framework and turning on risk assessment.

SESSION 11 - CORE - RESILIENCE BEYOND BOUNCE-BACK

```
RESILIENCE = the ability of a system, community or society exposed to
hazards to RESIST, ABSORB, ACCOMMODATE, ADAPT TO, TRANSFORM and RECOVER
from a hazard's effects in a timely and efficient manner, preserving and
restoring essential structures and functions

FRAMEWORK ELEMENTS AN ANSWER MUST NAME
  RISK KNOWLEDGE    do we know the hazard, exposure, vulnerability
                    and capacity we actually face?
  ROBUSTNESS        can the system take the shock without major
                    functional loss?
  REDUNDANCY        are there backup routes, supplies and systems?
  RESOURCEFULNESS   can resources be mobilised and reallocated under
                    stress?
  RAPID RECOVERY    can function be restored quickly and more safely?
  ADAPTIVE CAPACITY can the system change its form when the old form
                    no longer works?

TRAP: "bouncing back" restores the condition that failed
```

Six capabilities, not one adjective: the framework elements are what a resilience question is marked on.

DEFINITION / WHAT THIS IS CALLED

Plain definition: Resilience is the ability to take a hit, keep working, and come back stronger rather than merely the same.

Technical definition: Resilience is the capacity of an exposed system, community or society to resist, absorb, accommodate to, adapt to, transform and recover from the effects of a hazard in a timely and efficient manner, including through the preservation and restoration of essential basic structures and functions.

ANSWER-GRABBING OPENING - WRITE/ADAPT IN THE EXAM

> Resilience is not the ability to return to the condition that failed; it is the ability to keep functioning through a shock and to emerge with less risk than before.

MUST-WRITE KEYWORDS

resilience | **robustness** | **redundancy** | **resourcefulness** | **rapid recovery** | **adaptive capacity**

CORE EXPLANATION

The definition is built from six verbs, and dropping the later ones is the standard error. Resist and absorb describe withstanding the shock; accommodate describes continuing to function in a degraded but viable state; adapt and transform describe changing the system so that the same shock does less harm next time; recover describes restoring essential structures and functions. The timeliness and efficiency clause matters because a system that eventually recovers after a decade has not demonstrated resilience.

Resilience is determined jointly by the four risk components. For a given hazard and exposure, lower vulnerability and higher capacity produce higher resilience, which is why the determination limb of any resilience question should be answered through hazard, exposure, vulnerability and capacity rather than through a list of virtues. The framework elements then translate that determination into assessable features: risk knowledge, robustness, redundancy, resourcefulness, rapid recovery and adaptive capacity.

WORKED EXAMPLE

A district hospital with an engineered structure (robustness), a second power and water source (redundancy), an updated hazard and casualty assessment (risk knowledge), a surge staffing roster (resourcefulness), a service-restoration protocol (rapid recovery) and a plan to convert wards for a different hazard profile (adaptive capacity) is resilient. A hospital that merely survives collapse while losing power, water and staff is robust but not resilient.

TECHNICAL DISTINCTION

Capacity and resilience are not the same. Capacity is the stock of strengths and resources available; resilience is the demonstrated performance of a system under stress across time. High capacity on paper can coexist with low resilience if coordination, information or maintenance fails during the event.

NAMED-EVIDENCE PARAGRAPH DEMONSTRATION

Claim: Resilience must be measured against named elements, because an unqualified claim that a system is "resilient" carries no evidentiary content. **Named evidence:** The standard treatment builds resilience from the disaster management cycle and DRR practice as the capacity to resist, absorb, accommodate and recover in a timely and efficient manner, and expressly warns that a complete resilience framework requires naming risk knowledge, robustness, redundancy, resourcefulness, rapid recovery and the capacity to adapt or transform. **Analysis:** Naming the elements converts resilience into an assessable checklist against which a specific hospital, grid or settlement can be tested, which is exactly what a "how is it determined" demand asks for. **Qualification:** Measuring resilience additionally requires data on capacity, not merely on hazard and vulnerability, and the source vocabulary does not supply named indicators, so a resilience claim should be stated with its metric or not at all.

EXAMINER CAUTION

- Do not define resilience as bouncing back; the adapt and transform verbs carry the marks.
- Do not assert that a system is resilient without naming the element and the evidence behind the claim.

EXAM LINK

- **Prelims:** Options that omit adaptation and transformation from the resilience set are distractors.
- **Mains:** The 2024 GS-III question is answered by moving from definition to determination to framework elements in that order.

MINI RECAP

- Six verbs: resist, absorb, accommodate, adapt, transform, recover.
- Six elements: risk knowledge, robustness, redundancy, resourcefulness, rapid recovery, adaptive capacity.

SESSION 12 - CORE - INDIA'S POLICY BRIDGE: DM ACT 2005, NPDM 2009 AND THE PM's TEN-POINT AGENDA

```
2005  DISASTER MANAGEMENT ACT
      statutory basis for the national disaster-management system
        |
2009  NATIONAL POLICY ON DISASTER MANAGEMENT
      policy shift from a relief-centric to a prevention,
      mitigation and preparedness-centric approach
        |
------- 18 MARCH 2015: SENDAI FRAMEWORK ADOPTED -------------------
        |
2016  NATIONAL DISASTER MANAGEMENT PLAN (first edition)
      Sendai priorities carried into national planning
        |
Nov   PM's TEN-POINT AGENDA ON DISASTER RISK REDUCTION
2016  announced at the Asian Ministerial Conference on Disaster
      Risk Reduction (AMCDRR), NEW DELHI
      each point aligned with one or more Sendai priorities of action
        |
2019  REVISED NATIONAL DISASTER MANAGEMENT PLAN
      latest published NDMP; cite as the operative plan
```

The timeline supplies the dated "before and after Sendai" anchors that a comparison question requires.

DEFINITION / WHAT THIS IS CALLED

Plain definition: India built its disaster-management system before Sendai and then aligned its plans and political commitments with Sendai afterwards.

Technical definition: The Disaster Management Act, 2005 and the National Policy on Disaster Management, 2009 are India's pre-Sendai institutional instruments; the National Disaster Management Plan, 2016 (revised 2019) and the Prime Minister's ten-point agenda on disaster risk reduction, announced at AMCDRR in New Delhi in November 2016 with each point aligned to one or more Sendai priorities of action, are the principal post-Sendai instruments.

ANSWER-GRABBING OPENING - WRITE/ADAPT IN THE EXAM

> India's post-Sendai contribution was not to adopt a global vocabulary but to translate it: the ten-point agenda maps political commitments point by point onto Sendai's priorities of action.

MUST-WRITE KEYWORDS

Disaster Management Act, 2005 | National Policy on Disaster Management, 2009 | ten-point agenda | AMCDRR | New Delhi, November 2016 | Sendai alignment

CORE EXPLANATION

The 2005 Act created the statutory architecture within which every later plan operates, and the 2009 Policy articulated the shift away from a purely relief-centric approach towards prevention, mitigation and preparedness. Both precede Sendai, which is why a "before and after" question can be answered with dated Indian instruments rather than with a vague claim that India had no system before 2015.

The ten-point agenda, announced at the Asian Ministerial Conference on Disaster Risk Reduction held in New Delhi in November 2016, is the bridge. Its significance for an answer is structural rather than enumerative: each point is explicitly aligned with one or more Sendai priorities of action, so it demonstrates deliberate national translation of the global framework rather than incidental overlap. Its themes run from embedding disaster-risk management in all development sectors and encouraging risk coverage for the poorest through to greater involvement of women, investment in risk mapping, technology, university networks, social media, local capacity, learning from every disaster and stronger international cooperation on response.

WORKED EXAMPLE

Asked to show what India did after signing Sendai, an answer that cites only "India supports Sendai" earns nothing. An answer that cites the 2016 NDMP carrying Sendai's priorities into planning, the ten-point agenda announced at AMCDRR New Delhi in November 2016 with point-by-point Sendai alignment, and the 2019 revision of the NDMP, earns the evidence mark three times.

TECHNICAL DISTINCTION

The ten-point agenda is a political and programmatic statement of intent, not a statutory instrument. It neither amends the 2005 Act nor creates enforceable obligations, so it should be cited as evidence of alignment and direction rather than as a legal change.

NAMED-EVIDENCE PARAGRAPH DEMONSTRATION

Claim: India's alignment with Sendai is demonstrable from dated domestic instruments rather than asserted. **Named evidence:** The record of India's post-2015 policy shows that the Prime Minister's ten-point agenda on disaster risk reduction was announced at AMCDRR in New Delhi in November 2016 and that each of its points is explicitly aligned with one or more Sendai priorities of action, while the National Disaster Management Plan of 2016 incorporates the four Sendai priorities into its thematic structure. **Analysis:** Together these establish that India's post-2015 disaster policy was consciously mapped onto the global framework, which is the precise evidentiary content a "before and after signing" demand is testing. **Qualification:** Alignment of plans and agendas is an input; it does not by itself establish measured reduction in mortality, losses or exposure, for which separate dated evidence is required.

EXAMINER CAUTION

- Do not date the ten-point agenda to the Sendai conference; it was announced at AMCDRR in New Delhi in November 2016.
- Do not present the 2005 Act or 2009 Policy as responses to Sendai; both predate it.

EXAM LINK

- **Prelims:** The forum, city and year of the ten-point agenda are directly testable facts.
- **Mains:** These four dated instruments are the backbone of the "before and after Sendai" limb of the 2018 question.

MINI RECAP

- Before Sendai: DM Act 2005, NPDM 2009.
- After Sendai: NDMP 2016 revised 2019, and the ten-point agenda at AMCDRR New Delhi, November 2016.

SESSION 13 - CORE - NDMP: FIVE THEMATIC AREAS, THE 2016 CRITIQUE AND THE 2019 REVISION

```
NDMP 2016 - FIVE THEMATIC AREAS FOR ACTION
  (the four Sendai priorities carried into national planning)

  1  UNDERSTANDING RISK
  2  INTER-AGENCY COORDINATION
  3  INVESTING IN DRR - STRUCTURAL MEASURES
  4  INVESTING IN DRR - NON-STRUCTURAL MEASURES
  5  CAPACITY DEVELOPMENT

DOCUMENT-PERIOD CRITIQUE OF THE 2016 EDITION
  "too generic"                    no precise time frame
  no projected fund requirement    no goals or targets of its own
        (unlike Sendai and the Sustainable Development Goals)

CURRENT STATUS
  NDMP 2016 = first edition
  NDMP 2019 = latest published National Disaster Management Plan
              -> cite the 2019 revision as the operative plan
```

One plan, five thematic areas, four named weaknesses of its first edition, and one current-citation rule.

DEFINITION / WHAT THIS IS CALLED

Plain definition: India's national plan organises disaster work under five themes, and the first edition was criticised for having no targets of its own.

Technical definition: The National Disaster Management Plan, 2016 incorporates the four Sendai priorities into five Thematic Areas for Action - Understanding Risk, Inter-Agency Coordination, Investing in DRR through structural measures, Investing in DRR through non-structural measures, and Capacity Development; the revised NDMP of 2019 is the latest published edition.

ANSWER-GRABBING OPENING - WRITE/ADAPT IN THE EXAM

> The NDMP's first edition reproduced Sendai's priorities without reproducing Sendai's discipline of measurable targets, which is the sharpest available illustration of the difference between a framework and an accountability mechanism.

MUST-WRITE KEYWORDS

Understanding Risk | **Inter-Agency Coordination** | **structural investment** | **non-structural investment** | **Capacity Development** | **NDMP 2019**

CORE EXPLANATION

The five thematic areas are a national restatement of the Sendai priorities, with Sendai's third priority on investment split into structural and non-structural limbs. Understanding Risk corresponds to Sendai Priority 1 and covers hazard, vulnerability and capacity assessment. Inter-Agency Coordination corresponds to Priority 2 and covers the governance machinery across the Union, States and districts. The two investment themes correspond to Priority 3 and separate physical works from regulation, planning, insurance and awareness. Capacity Development covers training, institutional strengthening and community capability.

The recorded critique of the 2016 edition is equally examinable. It was described as too generic, without a precise time frame, without a projection of fund requirements, and - unlike Sendai or the Sustainable Development Goals - without goals or targets of its own. That last point is the analytical core: a plan that adopts a target-driven framework's priorities but not its targets cannot be monitored in the way the framework it copies can be. The correct current-citation rule is that the 2016 document was the first edition, and the revised NDMP of 2019 is the latest published plan.

WORKED EXAMPLE

Asked whether India's plan matches the global standard, a strong answer states that the NDMP mirrors Sendai's priorities thematically, then notes that the 2016 edition set no goals, targets, timeline or costed fund requirement, and concludes that monitoring therefore has to rely on scheme-level reporting rather than on plan-level indicators - while citing the 2019 revision as the operative plan.

TECHNICAL DISTINCTION

A thematic area is an organising heading; a target is a measurable commitment with a deadline. Sendai has both, and the 2016 NDMP had only the former, which is why "India has a plan aligned with Sendai" and "India has Sendai-equivalent accountability" are different claims.

NAMED-EVIDENCE PARAGRAPH DEMONSTRATION

Claim: The 2016 NDMP illustrates a framework that names priorities without a fully specified accountability mechanism. **Named evidence:** The plan sets out five Thematic Areas for Action, while the source assessment characterises the 2016 edition as too generic, lacking a precise time frame and projected fund requirements and, unlike Sendai or the Sustainable Development Goals, lacking its own goals or targets. **Analysis:** Because the themes were adopted from a target-bearing framework while the targets were not, activity could be reported without an equivalent results framework, which is the structural weakness a critical question is testing. **Qualification:** This is an external assessment of the first edition, not the plan's self-description; whether the 2019 revision closed these gaps requires a dated comparison and should not be assumed.

EXAMINER CAUTION

- Do not cite the 2016 edition as the current plan; the 2019 revision is the latest published NDMP.
- Do not carry the 2016 critique forward to the 2019 edition without a dated comparison.

EXAM LINK

- **Prelims:** The five thematic areas and the 2016-versus-2019 edition distinction are both testable.
- **Mains:** The 2016 critique is the domestic half of any answer contrasting India's plan with Sendai's quantified targets.

MINI RECAP

- Five themes: understanding risk, coordination, structural investment, non-structural investment, capacity development.
- 2016 first edition criticised for no timeline, no costing and no targets; 2019 revision is the operative plan.

SESSION 14 - CORE SYNTHESIS - FROM HYOGO TO SENDAI: ADOPTION, COMPARISON AND THE FOUR PRIORITIES

	HYOGO FRAMEWORK FOR ACTION 2005-2015	SENDAI FRAMEWORK FOR DRR 2015-2030
Adopted	World Conference on DRR, Kobe, Hyogo, Japan, 2005	Third UN World Conference on DRR, Sendai, Japan, 18 MARCH 2015
Structure	FIVE priorities	FOUR priorities
Measurable targets	none	SEVEN global targets
Emphasis	building resilience of nations and communities	preventing new risk and reducing existing risk
Legal character	voluntary, non-binding	voluntary, non-binding

SENDAI'S FOUR PRIORITIES FOR ACTION

1 UNDERSTANDING DISASTER RISK
2 STRENGTHENING DISASTER RISK GOVERNANCE TO MANAGE DISASTER RISK
3 INVESTING IN DISASTER RISK REDUCTION FOR RESILIENCE
4 ENHANCING DISASTER PREPAREDNESS FOR EFFECTIVE RESPONSE AND TO "BUILD BACK BETTER" IN RECOVERY, REHABILITATION AND RECONSTRUCTION

The comparison matrix answers the second limb of the 2018 question in a single glance.

DEFINITION / WHAT THIS IS CALLED

Plain definition: Sendai is the successor to Hyogo: fewer priorities, but for the first time, measurable global targets.

Technical definition: The Sendai Framework for Disaster Risk Reduction 2015-2030 was adopted at the Third UN World Conference on Disaster Risk Reduction in Sendai, Japan on 18 March 2015 as the voluntary, non-binding successor to the Hyogo Framework for Action 2005-2015, and is described as the first major agreement of the

post-2015 development agenda, with four priorities for action and seven global targets.

ANSWER-GRABBING OPENING - WRITE/ADAPT IN THE EXAM

> Sendai's real innovation over Hyogo was not a shorter list of priorities but the arrival of measurable targets in a field that had until then been governed entirely by exhortation.

MUST-WRITE KEYWORDS

Hyogo Framework for Action 2005-2015 | five priorities | 18 March 2015 | four priorities | Third UN World Conference on DRR | voluntary and non-binding

CORE EXPLANATION

Hyogo, adopted in 2005 in the aftermath of the 2004 Indian Ocean tsunami, organised global action under five priorities and set no quantified global targets, so progress could be described but not measured against a common yardstick. Sendai replaced it with four priorities that run in a deliberate logical order: understand the risk first; govern it through coherent institutions and coordination; invest in reducing it, treating disaster risk reduction as an investment rather than a residual cost; and prepare to respond and to Build Back Better in recovery, rehabilitation and reconstruction. Sendai also widened the scope beyond natural hazards to include man-made, technological and biological hazards, and shifted the emphasis from managing disasters to managing disaster risk.

Sendai was adopted on 18 March 2015 and was the first of the landmark 2015 agreements, preceding the 2030 Agenda for Sustainable Development adopted in September 2015 and the Paris Agreement adopted in December 2015. Sendai's own text therefore could not have recognised agreements that had not yet been concluded. The coherence between disaster risk reduction, sustainable development and climate action was developed subsequently, and it is later UNDRR material - including the 2023 Midterm Review - that stresses this coherence as a matter of implementation.

WORKED EXAMPLE

Asked how Sendai differs from Hyogo, a scoring answer states the count difference (five priorities against four), the substantive addition (seven measurable global targets with baselines and deadlines), the scope widening (from natural hazards to man-made, technological and biological hazards) and the constant (both are voluntary and non-binding), rather than simply asserting that Sendai is "more comprehensive".

TECHNICAL DISTINCTION

Priorities and targets are different instruments. A priority states what action to take; a target states what result to achieve by when. Reciting the four priorities in answer to a targets question, or the reverse, is the single most common error on this topic.

NAMED-EVIDENCE PARAGRAPH DEMONSTRATION

Claim: The Hyogo-to-Sendai transition is best described as the arrival of measurability, not of ambition. **Named evidence:** The framework record shows that Sendai was adopted at the Third UN World Conference on Disaster Risk Reduction in Sendai, Japan in 2015 as the successor to the Hyogo Framework for Action 2005-2015, that Hyogo carried five priorities, and that Sendai carries four priorities and seven global targets. **Analysis:** Because Hyogo's five priorities were not attached to quantified outcomes, national reporting under it could not be compared across countries, whereas Sendai's lettered targets create a common monitoring frame even though compliance remains voluntary. **Qualification:** Both frameworks are voluntary and non-binding, so the arrival of targets improves measurability without creating enforceability.

EXAMINER CAUTION

- Do not describe Sendai as a binding treaty; it is voluntary and non-binding.
- Do not claim that Sendai explicitly recognises the Paris Agreement or the 2030 Agenda; both were adopted later in 2015, and coherence was developed afterwards.

EXAM LINK

- **Prelims:** The adoption date, the venue, the five-versus-four priority counts and the targets distinction are all testable.

- **Mains:** This matrix is the second limb of the 2018 question and should be written as a comparison, not as two separate descriptions.

MINI RECAP

- Hyogo 2005-2015: five priorities, no targets. Sendai 2015-2030: four priorities, seven targets, adopted 18 March 2015.
- Both voluntary; Sendai was the first of the landmark 2015 agreements.

SESSION 15 - CORE SYNTHESIS - THE SEVEN GLOBAL TARGETS, EW4All AND THE 2023 MIDTERM REVIEW

A-D LOSS-REDUCTION OUTCOMES	E-G MEANS OF IMPLEMENTATION
A substantially reduce global disaster MORTALITY by 2030	E substantially increase the number of countries with national and local DRR STRATEGIES by 2020 -> the only target with a 2020 date
B substantially reduce the number of AFFECTED PEOPLE globally by 2030 (A and B measured as per-100,000 decadal averages, 2020-2030 against 2005-2015)	F substantially enhance INTERNATIONAL COOPERATION to developing countries by 2030
C reduce direct disaster ECONOMIC LOSS relative to global GDP by 2030	G substantially increase availability of and access to MULTI-HAZARD EARLY WARNING SYSTEMS and disaster risk information and assessments by 2030
D substantially reduce damage to CRITICAL INFRASTRUCTURE and disruption of basic services, including health and education, by 2030	EW4All: UN Secretary-General's call, March 2022; launched at COP27, November 2022; every person protected by an early warning system by 2027 pillars: risk knowledge (UNDRR); observation and forecasting (WMO); dissemination (ITU); preparedness and response (IFRC); WMO and UNDRR co-lead

MIDTERM REVIEW: High-Level Meeting 17-19 MAY 2023 -> Political Declaration issued as UN GENERAL ASSEMBLY RESOLUTION 77/289 (18 May 2023)

The A-D against E-G split is the structural point; Target E's 2020 date and the 2027 EW4All date are the two most-tested numbers.

DEFINITION / WHAT THIS IS CALLED

Plain definition: Sendai sets seven lettered targets: four about cutting losses, three about building the means to do so.

Technical definition: Sendai's seven global targets are A mortality, B affected people, C direct economic loss relative to global GDP, D damage to critical infrastructure and disruption of basic services, E national and local DRR strategies, F international cooperation to developing countries, and G availability of and access to multi-hazard early warning systems and disaster risk information; Targets A and B are measured as per-100,000 decadal averages for 2020-2030 against 2005-2015, and Target E alone carries a 2020 deadline.

ANSWER-GRABBING OPENING - WRITE/ADAPT IN THE EXAM

> Sendai's targets are structurally asymmetric: A-D measure loss outcomes no single state controls unilaterally, while E-G measure implementation conditions that depend on national action as well as international cooperation, technology and access.

MUST-WRITE KEYWORDS

Targets A-G | **loss-reduction targets** | **means-of-implementation targets** | **Target E 2020 deadline** | **EW4All 2027** | **UNGA resolution 77/289**

CORE EXPLANATION

Targets A to D are outcome targets. A and B address human loss - mortality and affected people - and are measured as per-100,000 decadal averages for 2020-2030 against the 2005-2015 baseline, which is why single-year comparisons are not valid evidence about them. C addresses direct economic loss expressed relative to global gross domestic product, so growth changes the denominator. D addresses damage to critical infrastructure and disruption of basic services including health and education, by developing their resilience.

Targets E to G address means of implementation. E asks countries to adopt national and local DRR strategies by 2020 and is the only target with an earlier deadline. F asks for enhanced international cooperation to developing countries. G asks for increased availability of and access to multi-hazard early warning systems and disaster risk information and assessments. Its operational vehicle is Early Warnings for All, called for by the UN Secretary-General in March 2022 and launched as an initiative at COP27 in November 2022, aiming to protect every person on earth with an early warning system by 2027 across four pillars - risk knowledge led by UNDRR, observation and forecasting by WMO, dissemination and communication by ITU, and preparedness and response by IFRC, with WMO and UNDRR co-leading overall.

Implementation status has its own dated anchor. The High-Level Meeting on the Midterm Review of the Sendai Framework was held on 17-19 May 2023 and adopted a Political Declaration, issued as UN General Assembly resolution 77/289 on 18 May 2023. Any statement about global progress against the seven targets must be sourced to that review rather than to the 2015 description of the targets themselves.

WORKED EXAMPLE

Asked to "mention the global targets", the correct move is to name all seven compactly with their letters and the 2020/2030 split, spending no more than a line each, rather than describing three at length and omitting four - the directive "mention" rewards completeness over depth.

TECHNICAL DISTINCTION

A global target is not a national indicator. Sendai's targets are aggregate global commitments; a country's own performance is reported through national indicators and is not established merely by quoting a global target. Equally, an initiative such as EW4All evidences how Target G is being pursued, not that warning coverage has been achieved.

NAMED-EVIDENCE PARAGRAPH DEMONSTRATION

Claim: Sendai separates loss outcomes from the means used to pursue them, but the means targets do not all sit within one state's direct control. **Named evidence:** Targets A-D measure mortality, affected people, economic loss and infrastructure disruption; Target E concerns national and local strategies, Target F international cooperation and Target G warning and risk-information access. **Analysis:** Target E is comparatively straightforward to verify as a domestic policy input, whereas F and G also depend on external cooperation, technology, finance and last-mile access; progress on means therefore cannot be equated with improved loss outcomes. **Qualification:** This is an analytical reading of the framework's design, not a ranking of targets by difficulty, and actual progress must be sourced to the 2023 Midterm Review rather than inferred from the target list.

EXAMINER CAUTION

- Do not attach a 2020 deadline to any target other than E.
- Do not use EW4All or a national strategy count as evidence that disaster losses have fallen.

EXAM LINK

- **Prelims:** Target lettering, the A-D against E-G split, Target E's 2020 deadline, the 2027 EW4All year and resolution 77/289 are all directly testable.
- **Mains:** The 2024 question's final limb is satisfied by naming all seven targets compactly with the deadline distinction.

MINI RECAP

- A-D reduce losses; E-G build means of implementation; E alone is a 2020 target.
- EW4All targets 2027 for Target G; the Midterm Review Political Declaration became UNGA resolution 77/289 on 18 May 2023.

BASIC MCQS / REMEDIATION

Q1. A severe earthquake strikes an uninhabited high-altitude desert and causes no casualties or asset loss. Which statement is correct?

A. It is a hazard event but not a disaster, because exposure was negligible and no community's coping capacity was exceeded. B. It is a disaster, because disaster classification depends on the magnitude of the geophysical process alone. C. It is neither a hazard nor a disaster, because hazards are defined only where people are present. D. It is a disaster only if a government later declares it one for administrative purposes.

Answer: A.

Option-by-option explanation:

- **A (correct):** The hazard occurred, but with negligible exposure the loss never crossed the coping-capacity threshold that defines a disaster.
- **B:** Magnitude describes the process; the definition turns on losses beyond a community's ability to cope.
- **C:** A hazard exists as a dangerous condition or event regardless of who is present.
- **D:** Administrative declaration is a procedural consequence, not the conceptual threshold.

Remediation: Re-read Session 1 and rehearse the sentence "a disaster is a threshold, not an event."

Q2. Which grouping correctly lists the natural hazard families recognised in the source classification?

A. Geophysical, industrial, chemical, biological and radiological. B. Geophysical, hydrological, climatological, meteorological and biological. C. Coastal, riverine, urban, rural and mountainous. D. Rapid, slow, compound, cascading and residual.

Answer: B.

Option-by-option explanation:

- **A:** Industrial, chemical and radiological hazards are anthropogenic, not natural families.
- **B (correct):** These are the five natural families; anthropogenic hazards form a separate class.
- **C:** These are settlement or terrain descriptors, not hazard families.
- **D:** These describe onset or interaction characteristics, not originating processes.

Remediation: Redraw the Session 2 hazard tree and place one Indian example under each of the five natural families.

Q3. 'Capacity', within the hazard-exposure-vulnerability-capacity framework, is best understood as:

A. The maximum discharge a river channel can carry before it overtops its banks. B. The number of people a State can evacuate within twenty-four hours. C. The combination of strengths, attributes, resources, institutions and measures available to manage and reduce disaster risk. D. The financial ceiling of the relief fund available to a district administration.

Answer: C.

Option-by-option explanation:

- **A:** This is a hydraulic property of a channel, not a risk-framework term.
- **B:** Evacuation throughput is one indicator of capacity, not its definition.
- **C (correct):** Capacity is the aggregate of strengths, resources, institutions and measures that contain harm.
- **D:** A financial ceiling is one resource among many and is post-event in character.

Remediation: List, for your own district, one institutional, one human and one financial element of capacity.

Q4. The working relation Risk = Probability of Hazard x Degree of Vulnerability should be used:

A. As a validated predictive formula that yields the expected number of deaths in any future event. B. As a legal standard against which compensation entitlements are computed. C. As a replacement for the four-component hazard-exposure-vulnerability-capacity framework. D. As a disciplined heuristic that organises the components of a risk problem and identifies which factor can actually be moved.

Answer: D.

Option-by-option explanation:

- **A:** The source offers it as a working relation, not as a validated predictive formula.
- **B:** Compensation is governed by administrative norms, not by this relation.
- **C:** It omits exposure and capacity as explicit terms and therefore supplements rather than replaces the four-component framework.
- **D (correct):** Its value lies in ordering the problem and locating the reducible factor.

Remediation: Write one sentence using the relation that names the area and the time period to which the risk claim applies.

Q5. Which option correctly lists the four ways of dealing with risk?

A. Risk acceptance, risk avoidance, risk reduction and risk transfer. B. Risk mapping, risk zoning, risk costing and risk auditing. C. Prevention, mitigation, preparedness and response. D. Relief, rehabilitation, reconstruction and recovery.

Answer: A.

Option-by-option explanation:

- **A (correct):** These are the four distinct risk-treatment strategies.
- **B:** These are risk-assessment activities, not treatment options.
- **C:** These are phases of the management cycle, not treatment choices for an identified risk.
- **D:** These are post-impact phases of the Crisis Management Phase.

Remediation: For one hazard in your notes, assign a concrete measure to each of the four options.

Q6. Which pairing correctly distinguishes rapid-onset from slow-onset hazards?

A. Drought and desertification are rapid onset; earthquakes and tsunamis are slow onset. B. Earthquake, cyclone, flood and tsunami are rapid onset; climate change, desertification, soil degradation and drought are slow-onset 'creeping emergencies'. C. All hydro-meteorological hazards are slow onset and all geophysical hazards are rapid onset. D. Onset speed is determined by the duration of the recovery period after the event.

Answer: B.

Option-by-option explanation:

- **A:** The categories are inverted.
- **B (correct):** This is the source classification, with creeping emergencies named for the slow-onset set.
- **C:** Floods and cyclones are hydro-meteorological and rapid onset, so the rule fails.
- **D:** Onset speed refers to the development of the hazard impact, not to recovery duration.

Remediation: State the loss mechanism of a creeping emergency in one line: progressive loss of the ability to derive sustenance.

Q7. Beyond physical, social, economic and environmental vulnerability, which further dimension does the five-fold typology add?

A. Meteorological vulnerability, reflecting seasonal hazard variability. B. Administrative vulnerability, reflecting district staffing levels. C. Attitudinal vulnerability, reflecting a community's negative disposition towards change and consequent dependence on external support. D. Fiscal vulnerability, reflecting the size of the State disaster response fund.

Answer: C.

Option-by-option explanation:

- **A:** Seasonal variability is a hazard characteristic, not a condition of the exposed community.
- **B:** Staffing is an element of institutional capacity.
- **C (correct):** Attitudinal vulnerability is the fifth named dimension and the one most often omitted.
- **D:** Fund size is a financing capacity variable.

Remediation: Add one line on attitudinal vulnerability to your standard vulnerability answer template.

Q8. Resilience, as defined for disaster management, includes all of the following EXCEPT:

A. Absorbing and accommodating the effects of a hazard while continuing to function. B. Adapting and transforming so that the same shock produces less harm in future. C. Restoring essential basic structures and functions in a timely and efficient manner. D. Returning the affected system to precisely its pre-disaster condition, including the conditions that produced the loss.

Answer: D.

Option-by-option explanation:

- **A:** Absorption and accommodation are explicit components of the definition.
- **B:** Adaptation and transformation are integral, not optional, components.
- **C:** Preservation and restoration of essential structures and functions is part of the definition.
- **D (correct):** Identical restoration reinstates the vulnerability that failed and is the trap the definition warns against.

Remediation: Recite the six verbs - resist, absorb, accommodate, adapt, transform, recover - before writing any resilience answer.

Q9. The replacement of the traditional '3 Rs' by the '3 Ps' signifies:

A. A shift from reactive Rescue, Relief and Restoration towards anticipatory Prevention, Preparedness and Proofing. B. A shift from Prevention, Preparedness and Proofing back to Rescue, Relief and Restoration after repeated implementation failures. C. A renaming of the same three activities with no change of policy emphasis. D. The replacement of State responsibility for relief with a wholly centralised response system.

Answer: A.

Option-by-option explanation:

- **A (correct):** The source records the replacement of the reactive triad by the anticipatory one.
- **B:** The direction of the shift is the reverse.
- **C:** Restoration and proofing are substantively different: one reinstates the prior state, the other makes it hazard-resistant.
- **D:** The shift concerns approach and sequencing, not the distribution of responsibility between the Union and the States.

Remediation: Write the two triads side by side and underline 'Restoration' against 'Proofing'.

Q10. The Prime Minister's ten-point agenda on disaster risk reduction was announced at:

A. The Third UN World Conference on Disaster Risk Reduction, Sendai, in March 2015. B. The Asian Ministerial Conference on Disaster Risk Reduction held in New Delhi in November 2016. C. The High-Level Meeting on the Sendai Midterm Review in May 2023. D. COP27 in November 2022.

Answer: B.

Option-by-option explanation:

- **A:** Sendai 2015 was where the framework was adopted, not where the ten-point agenda was announced.
- **B (correct):** The agenda was announced at AMCDRR in New Delhi in November 2016, with each point aligned to Sendai priorities.
- **C:** The 2023 meeting adopted the Midterm Review Political Declaration.
- **D:** COP27 launched the Early Warnings for All initiative.

Remediation: Fix the triple - forum AMCDRR, city New Delhi, month and year November 2016.

Q11. The Sendai Framework for Disaster Risk Reduction 2015-2030 was adopted:

A. At the World Conference on Disaster Reduction in Kobe, Hyogo, in January 2005. B. By a binding protocol to the United Nations Framework Convention on Climate Change. C. At the Third UN World Conference on Disaster Risk Reduction in Sendai, Japan, on 18 March 2015, as the voluntary successor to the Hyogo Framework for Action. D. By the UN General Assembly in resolution 77/289 of 18 May 2023.

Answer: C.

Option-by-option explanation:

- **A:** That conference adopted the Hyogo Framework for Action 2005-2015.
- **B:** Sendai is not a climate-treaty instrument and is not binding.
- **C (correct):** This states the conference, venue, date and legal character correctly.
- **D:** Resolution 77/289 issued the 2023 Midterm Review Political Declaration, long after adoption.

Remediation: Memorise the date 18 March 2015 together with the phrase "voluntary and non-binding".

Q12. Compared with the Hyogo Framework for Action 2005-2015, the Sendai Framework:

A. Increased the number of priorities from four to five and removed all quantified targets. B. Converted a voluntary framework into a legally enforceable treaty with compliance penalties. C. Retained Hyogo's five priorities unchanged while adding a financing mechanism. D. Reorganised action under four priorities instead of five and, for the first time, added seven measurable global targets while remaining voluntary.

Answer: D.

Option-by-option explanation:

- **A:** The priority count moved from five to four, and targets were added rather than removed.
- **B:** Sendai remains voluntary and non-binding.
- **C:** The priorities were reorganised, not retained.
- **D (correct):** Fewer priorities, seven new measurable targets, unchanged voluntary character.

Remediation: Rehearse the four-cell comparison: priority count, targets, scope, legal character.

Q13. Which option correctly lists Sendai's four priorities for action?

A. Understanding disaster risk; strengthening disaster risk governance; investing in disaster risk reduction for resilience; enhancing preparedness for effective response and to Build Back Better in recovery, rehabilitation and reconstruction. B. Rescue; relief; restoration; rehabilitation. C. Mortality reduction; economic loss reduction; infrastructure protection; early warning. D. Understanding risk; inter-agency coordination; structural investment; non-structural investment.

Answer: A.

Option-by-option explanation:

- **A (correct):** This is the exact four-priority sequence.
- **B:** These are phases of a reactive response model, not Sendai priorities.
- **C:** These are target subjects, not priorities.
- **D:** These are four of the National Disaster Management Plan's five thematic areas.

Remediation: Learn the priorities in their logical order - understand, govern, invest, prepare and build back better.

Q14. Which statement correctly describes the internal structure of Sendai's seven global targets?

A. All seven targets measure loss outcomes and carry identical 2030 deadlines. B. Targets A-D measure loss-reduction outcomes, while Targets E-G measure means of implementation such as strategies, cooperation and early warning. C. Targets A-C are national indicators and Targets D-G are global indicators. D. Targets A-G are ranked in descending order of legal enforceability.

Answer: B.

Option-by-option explanation:

- **A:** E-G are means-of-implementation targets and Target E carries a 2020 deadline.
- **B (correct):** This is the structural split that an analytical answer should state.
- **C:** All seven are global targets; national indicators are a separate reporting layer.
- **D:** None of the targets is legally enforceable; the framework is voluntary throughout.

Remediation: Write the A-D and E-G columns from memory and mark which column a state actually controls.

Q15. The Early Warnings for All initiative, aiming at universal early-warning coverage by 2027, is organised around four pillars led by:

A. UNDP, UNEP, UNESCO and UNICEF. B. The World Bank, IMF, WTO and UNCTAD. C. UNDRR for risk knowledge, WMO for observation and forecasting, ITU for dissemination and communication, and IFRC for preparedness and response, with WMO and UNDRR co-leading. D. NDMA, IMD, NRSC and NDRF.

Answer: C.

Option-by-option explanation:

- **A:** These agencies have important development mandates but are not the named EW4All pillar leads.
- **B:** These are economic and trade institutions.
- **C (correct):** These are the four pillars and their lead agencies, with WMO and UNDRR co-leading overall.
- **D:** These are Indian national institutions, not the global pillar leads.

Remediation: Map the four pillars onto the warning chain: know, detect, tell, act.

Q16. Which statement about India's National Disaster Management Plan is correct?

A. The NDMP 2016 is the only edition ever published and remains the sole operative plan. B. The NDMP 2016 set quantified national targets equivalent to Sendai's seven global targets. C. The NDMP is a statutory treaty obligation arising directly from the Sendai Framework. D. The NDMP 2016 organised the four Sendai priorities into five thematic areas but was criticised as too generic, without a precise time frame, fund projection or targets of its own, and the revised NDMP of 2019 is the latest published edition.

Answer: D.

Option-by-option explanation:

- **A:** The 2016 document was the first edition; a revised plan was published in 2019.
- **B:** The recorded critique is precisely that it set no goals or targets of its own.
- **C:** Sendai is voluntary; the NDMP derives from the Disaster Management Act, 2005.
- **D (correct):** This states the thematic structure, the document-period critique and the current edition correctly.

Remediation: Always pair the five thematic areas with the four-part critique and the 2019 citation rule.

PYQS AND ANSWER PRACTICE

The three questions below are reproduced exactly as set. Each carries a model answer written to the stated word limit and a checklist that converts the model into a repeatable answer route. Alternative model answers for the same three questions, together with six original Mains questions, are placed in the Solved Practice Workbook so that the two documents drill different things.

DIRECT PYQ - 2018 GS-III Q18 - 15 marks, 250 words

Question: Describe various measures taken in India for Disaster Risk Reduction (DRR) before and after signing 'Sendai Framework for DRR (2015-2030)'. How is this framework different from 'Hyogo Framework for Action, 2005'?

Model answer:

India's disaster risk reduction record falls into three blocks: measures before Sendai, measures after it, and how Sendai differs from Hyogo.

Before signing Sendai, India had already moved away from a purely relief-centric system. The Disaster Management Act, 2005 created the statutory architecture of authorities, plans and dedicated funds at national, State and district levels. The National Policy on Disaster Management, 2009 consolidated the reorientation towards prevention, mitigation and preparedness, replacing the reactive 3 Rs of Rescue, Relief and Restoration with the 3 Ps of Prevention, Preparedness and Proofing. Within this framework India developed hazard zoning, building codes, structural mitigation works and dedicated response forces.

After signing Sendai, India translated the framework domestically. The National Disaster Management Plan, 2016 carried Sendai's four priorities into five thematic areas - understanding risk, inter-agency coordination, structural investment in DRR, non-structural investment in DRR and capacity development - and was revised in 2019, which is the latest published edition. In November 2016 the Prime Minister announced a ten-point agenda on disaster risk reduction at the Asian Ministerial Conference on Disaster Risk Reduction in New Delhi, each point aligned with one or more Sendai priorities of action.

Sendai differs from Hyogo structurally rather than legally. Hyogo (2005-2015) organised action under five priorities without quantified global targets. Sendai, adopted on 18 March 2015, reorganises action under four priorities, adds seven measurable global targets, widens scope to man-made, technological and biological hazards, and shifts emphasis from managing disasters to managing disaster risk. Both remain voluntary and non-binding.

Answer checklist:

- Name the two pre-Sendai instruments with their years: DM Act 2005, NPDM 2009.
- Name the two post-Sendai instruments: NDMP 2016 revised 2019, and the ten-point agenda at AMCDRR New Delhi, November 2016.
- State the priority-count difference (five against four) and the target difference (none against seven).

- Add the scope widening and the managing-disasters to managing-disaster-risk shift.
- Close by noting that both frameworks are voluntary, so the change is in measurability, not enforceability.

DIRECT PYQ - 2019 GS-III - 10 marks, 150 words

Question: Vulnerability is an essential element for defining disaster impacts and its threat to people. How and in what ways can vulnerability to disasters be characterized? Discuss different types of vulnerability with reference to disasters.

Model answer:

Vulnerability is the set of physical, social, economic, environmental and attitudinal conditions that increase susceptibility to a hazard's impact. It explains why an identical hazard harms groups unequally. Economic vulnerability arises from narrow income diversity and weak access to production means. Physical vulnerability arises from proximity and construction type. Social vulnerability concentrates among the poor, pregnant and lactating women, persons with disability, children and the elderly. Environmental vulnerability reflects ecosystem sensitivity, such as cleared mangroves no longer buffering surge. Attitudinal vulnerability reflects resistance to change and dependence on external support. Where dimensions coincide, harm compounds. Vulnerability is policy-responsive, but it is not the only movable risk component: exposure can be reduced through siting and capacity strengthened through institutions, resources and preparedness.

Answer checklist:

- Define vulnerability and immediately state its explanatory role before listing anything.
- Use all five dimensions; the attitudinal dimension is the differentiator.
- Attach one concrete illustration to at least two dimensions within the word limit.
- State the compounding effect when dimensions coincide.
- Close on reducibility, which is the policy-relevant point at 10 marks.

DIRECT PYQ - 2024 GS-III Q17 - 15 marks, 250 words

Question: What is disaster resilience? How is it determined? Describe various elements of a resilience framework. Also mention the global targets of Sendai Framework for Disaster Risk Reduction (2015-2030).

Model answer:

Disaster resilience is the capacity of a system, community or society exposed to hazards to resist, absorb, accommodate to, adapt to, transform and recover from a hazard's effects in a timely and efficient manner, including the preservation and restoration of essential basic structures and functions.

It is determined by the configuration of the four risk components. For a given hazard and level of exposure, resilience rises as vulnerability falls and as capacity - institutions, finance, warning systems, trained responders, diversified livelihoods and robust infrastructure - increases. It is therefore a performance of a system over time, not a fixed attribute of a place.

The elements of a resilience framework are risk knowledge, meaning accurate information on hazard, exposure, vulnerability and capacity; robustness, the ability to withstand stress without major functional loss; redundancy, the availability of backup systems and alternative pathways; resourcefulness, the ability to mobilise and reallocate resources under stress; rapid recovery, the restoration of essential functions quickly and more safely; and adaptive capacity, the ability to change form when the existing form no longer works.

Globally, the Sendai Framework 2015-2030 sets seven targets: A, substantially reduce disaster mortality; B, reduce the number of affected people; C, reduce direct economic loss relative to global GDP; D, reduce damage to critical infrastructure and disruption of basic services; E, increase countries with national and local DRR strategies by 2020, the only earlier deadline; F, enhance international cooperation to developing countries; and G, increase access to multi-hazard early warning systems and risk information.

Answer checklist:

- Treat the question as four limbs and give each its own paragraph or clear sentence block.
- Define resilience with all six verbs; do not write "bounce back".

- Answer "how is it determined" through hazard, exposure, vulnerability and capacity, not through a list of virtues.
- Name six framework elements, each with a one-line operative meaning.
- Obey the directive "mention" for the targets: name all seven compactly with the 2020/2030 distinction rather than describing three at length.

SIX ANSWER-WRITING DRILLS

1. Write the disaster definition in one sentence that includes the phrase "beyond a community's ability to cope", then name the intervention point it implies.
2. Convert the five-fold vulnerability typology into a 150-word paragraph with one Indian illustration per dimension.
3. Assign a concrete measure to each of risk acceptance, avoidance, reduction and transfer for a single named hazard.
4. Draw the two-phase cycle from memory and place the 3Rs and 3Ps triads correctly against it.
5. Write the Hyogo-Sendai comparison as four contrast sentences: priority count, targets, scope, legal character.
6. Name all seven Sendai targets in under sixty words, marking the A-D and E-G split and Target E's 2020 deadline.

OPTIONAL ADVANCED DEPTH - NOT REQUIRED FOR A CORE ANSWER

Everything needed for a complete answer to the 2018, 2019 and 2024 questions is already in the Basic session above. This block adds the risk-governance layer that lifts a correct answer into an evaluative one. It analyses how risk is produced, why frameworks under-deliver, and where evidence must stop.

The central proposition

Disaster risk in India is not simply a natural given. It is substantially co-produced by development choices - unregulated construction, floodplain encroachment and deforestation among them. Resilience therefore cannot be treated as a static defensive quality of a place. It has to be measured, governed and continuously reinvested in. The Sendai Framework's own architecture, with priorities that carry no binding force and targets that carry no automatic enforcement, is itself a case study in the limits of global risk governance.

This block supplies the governance and measurement layer only. It does not analyse any individual hazard operationally; earthquakes, floods, cyclones and the rest are analysed in their own topics, which apply this layer hazard by hazard.

The risk-creation loop

```
DEVELOPMENT CHOICE
(land use, construction, drainage, agriculture and industrial
 siting, decided without risk information)
        |
        v
RISK CREATION AND ACCUMULATION
(exposure and vulnerability rise even where hazard
 frequency is unchanged)
        |
        v
DISASTER EVENT
(the hazard interacts with accumulated risk)
        |
        v
RESPONSE AND RECOVERY CHOICE
(reconstruct as before, or Build Back Better)
        |
        v
FEEDBACK
(reconstruction either recreates the same risk or reduces
 it for the next cycle)
```

Urban flooding in areas built with little regard to natural topography, and landslides along roads and settlements cut into unstable slopes, are two expressions of the same underlying phenomenon: risk is frequently created by development decisions taken without risk information, rather than inherited from nature. A resilience answer that discusses only response capacity, and never the loop that manufactures the risk being responded to, is analytically

incomplete.

The DRR sequence is a dependency chain

The field of action runs from the policy and legal framework through risk assessment, risk awareness and plan implementation to early warning and the use of knowledge. The advanced point is that this is a sequence with feedback rather than a menu of independent options. Risk assessment sits early because everything downstream consumes its output: awareness campaigns communicate whatever the assessment found, plans implement against the risks it identified, and warning systems are calibrated to the hazards it mapped. Skipping or under-resourcing assessment therefore degrades every later stage while leaving the framework looking complete on paper.

The accountability parallel between the NDMP and Sendai

The recorded critique of the 2016 National Disaster Management Plan is that it was too generic, carried no precise time frame, projected no fund requirement and, unlike the Sendai Framework or the Sustainable Development Goals, set no goals or targets of its own. That is a textbook description of a governance framework that names priorities without building an accountability mechanism.

The structural point is that Sendai suffers the same defect one level up. It sets seven global targets but, being voluntary and non-binding, has no enforcement mechanism against any individual member state's performance. The accountability gap therefore recurs at both the domestic and the global level, which is why a critique of India's plan and a critique of the framework it aligns with are really the same argument applied twice. The current-citation rule still applies: the 2016 document was the first edition, the revised plan of 2019 is the latest published edition, and whether the 2019 revision closed these gaps requires a dated comparison rather than an assumption in either direction.

The asymmetry between Targets A-D and Targets E-G

Sendai's internal structure carries a sharper problem than the absence of enforcement. Targets A to D measure loss outcomes - mortality, affected people, economic loss relative to global GDP, and damage to critical infrastructure - which no state controls unilaterally, since outcomes depend on hazard occurrence, global economic conditions and events elsewhere. Targets E to G measure means of implementation - national and local strategies, international cooperation, and warning coverage. Target E is directly attributable to domestic policy, whereas F and G also depend on other countries, international organisations, technology, finance and last-mile access.

Target E, on national and local DRR strategies, alone carried a 2020 deadline and is comparatively straightforward to verify as a domestic policy input. The qualified implication is that strategy adoption can improve while loss outcomes remain unchanged; this does not make E inherently less demanding, nor does it make cooperation and warning access directly controllable by one State.

The 2023 Midterm Review made the weakness official

The High-Level Meeting on the Midterm Review of the Sendai Framework was held on 17-19 May 2023 and adopted a Political Declaration, issued as United Nations General Assembly resolution 77/289 on 18 May 2023. Its significance for an answer is its character: it is a renewed political commitment rather than a binding correction, which is precisely the evidence needed to argue that Sendai's accountability deficit was acknowledged rather than remedied.

The concrete operational response of the same period was Early Warnings for All, called for by the UN Secretary-General in March 2022 and launched at COP27 in November 2022, with a 2027 horizon and four pillars led by UNDRR, WMO, ITU and IFRC. That response advanced a means target, Target G, rather than any of the loss-reduction targets - which reinforces rather than contradicts the asymmetry argument above.

On chronology, note that Sendai was adopted on 18 March 2015, before the 2030 Agenda for Sustainable Development in September 2015 and before the Paris Agreement in December 2015. Sendai's text could not have recognised agreements not yet adopted, and it is later UNDRR material, including the Midterm Review, that presses coherence between disaster risk reduction, sustainable development and climate action as an implementation requirement.

Two under-used analytical assets

The first is attitudinal vulnerability. The standard typology names a community's negative disposition towards change, and the dependence on external support it produces, as a distinct dimension alongside the economic, physical, social and environmental ones. Most answers stop at the first four, so citing the fifth signals deeper engagement with the source - provided it is argued as a product of past experience, exclusion and unmet promises rather than as a fault of the affected community.

The second is the slow-onset gap. The source is explicit that creeping emergencies such as climate change, desertification, soil degradation and drought must find adequate reflection in disaster preparedness, because societies lose their ability to derive sustenance over time even without a visible shock. India's institutional response, however, has historically been constructed around rapid-onset events with a declarable impact moment. That mismatch between stated importance and institutional design is a legitimate analytical point for any critically-examine demand.

The resilience measurement gap

Measuring resilience requires data on capacity, not merely on hazard and vulnerability. The working vocabulary of the field implies capacity as the denominator of risk, but it does not operationalise it with named indicators. The honest advanced position is therefore to state the measurement gap explicitly rather than to assert that resilience is high or low without a stated metric. The same discipline applies to progress claims: any statement about current global or Indian progress against the seven targets must rest on the 2023 Midterm Review material rather than on the 2015-era description of the targets, because target progress is time-variant.

Equity, trade-offs and the limits of framework existence

The five-fold vulnerability typology is the analytical base for equity-sensitive risk assessment. Where several dimensions coincide in the same population - an economically poor, socially marginal, physically exposed household - the resulting risk is compounding rather than merely additive, which is the technical reason targeted protection outperforms uniform protection.

Prevention and mitigation investment against response readiness is a genuine resource trade-off, not merely a sequencing preference. Money spent on structural risk reduction such as embankments and retrofitting competes with money spent on response capacity such as force strength and shelters within the same disaster finance envelope. Sendai's third priority is an explicit call to shift that balance towards prevention, yet the critique of the 2016 national plan - no costed fund requirement - shows that India's domestic plan did not operationalise the shift with a specific investment commitment.

That leads to the fallacy an advanced answer should name directly: the belief that because a framework exists, risk is being managed. Having a management cycle, a field of action and a national policy does not itself reduce risk. A framework can exist while lacking the timelines, funding and targets that make accountability possible. Distinguishing framework existence from framework effectiveness is the move that converts description into evaluation. A related discipline applies to Sendai's four priorities: they are diagnostic categories, not a checklist to recite, and an answer that identifies which priority is the binding constraint in the specific context asked about outperforms one that lists all four generically.

Monitoring, data and the evidence boundary

Monitoring the seven targets requires standardised national data on mortality, affected persons, economic loss, critical infrastructure damage, national and local strategies, international cooperation and early-warning access. That data architecture is a substantial requirement in its own right and is not supplied by the framework text. Consequently, any numeric claim about India's progress against a specific Sendai target requires a dated national or UNDRR source; the 2015-period description of the targets cannot support a current-progress claim.

Applying the advanced layer to the questions

For the 2024 resilience question, the advanced answer goes beyond the definitional response in three ways: it names the risk-creation loop as the reason resilience must be actively invested in rather than passively assumed; it notes the structural parallel between the 2016 plan's lack of targets and Sendai's non-binding character as a governance-design point; and it cites the 2023 Midterm Review for any claim about current global progress instead of reciting the 2015 target list as if it were a scoreboard.

For the 2018 question, the advanced layer adds the observation that the before-and-after comparison is a comparison of instruments, not of outcomes: dated Acts, policies, plans and agendas establish alignment and direction, while measured reduction in mortality, losses or exposure would require separate outcome evidence. For the 2019 question, it adds the attitudinal dimension and the compounding argument, and resists the temptation to treat vulnerability as a fixed characteristic of poor communities rather than as a produced condition.

Advanced answer spine

1. Define resilience precisely as the capacity to resist, absorb, accommodate, adapt, transform and recover, not as returning to normal.
2. Name the framework elements: risk knowledge, robustness, redundancy, resourcefulness, rapid recovery and adaptive capacity.
3. Show the risk-creation loop, so that resilience is presented as contending with actively produced risk and not only with natural hazard frequency.
4. Cite Sendai's four priorities and seven targets, and its status as reviewed in 2023.
5. Diagnose the domestic accountability gap using the recorded critique of the 2016 plan as evidence.
6. Recommend measurable, financed and time-bound targets as the corrective that directly addresses the gap identified.
7. Close by tying resilience back to risk-informed development through the feedback loop, rather than ending on relief and response.

CONSOLIDATED REGISTER NOTES

Risk grammar and the disaster threshold

1. **Hazard:** a dangerous condition or event that threatens, or has the potential for causing, injury to life or damage to property or the environment.
2. **Exposure:** the people, livelihoods, infrastructure and assets located where a hazard can reach them.
3. **Vulnerability:** conditions determined by physical, social, economic and environmental factors or processes that increase susceptibility to hazard impact.
4. **Capacity:** the combination of strengths, resources, institutions and measures available to manage and reduce disaster risk.
5. **Disaster:** serious disruption to the functioning of a community causing human, material, economic and environmental losses beyond the community's ability to cope, arising from hazards, conditions of vulnerability and insufficient capacity or risk-reduction measures.
6. **Threshold rule:** identical hazard magnitude produces a non-event or a disaster depending entirely on exposure, vulnerability and capacity.

Hazard classification and onset speed

7. **By source:** natural, socio-natural and anthropogenic.
8. **Natural families:** geophysical, hydrological, climatological, meteorological and biological.
9. **Anthropogenic examples:** pollution, deforestation, chemical spillage, industrial and transport accidents.
10. **Socio-natural:** a natural process whose severity is manufactured by human decisions, such as urban flooding after drainage encroachment.
11. **Rapid onset:** earthquake, cyclone, flood, tsunami - sudden, acute impact with a definable impact moment.
12. **Slow onset / creeping emergencies:** climate change, desertification, soil degradation, drought - unfold over months or years, societies progressively lose the ability to derive sustenance, and they must find adequate reflection in preparedness planning.

The five-fold vulnerability typology

13. **Economic:** income diversity and access to the means of production.
14. **Physical:** proximity to the hazard and type of construction.
15. **Social:** the poor, pregnant and lactating women, persons with disability, children, the elderly.
16. **Environmental:** ecosystem sensitivity and loss of natural buffers.
17. **Attitudinal:** a community's negative disposition towards change, producing dependence on external support - the dimension most often omitted and the one that earns the mark.
18. **Compounding rule:** where dimensions coincide in one population, risk compounds rather than adds.

Risk and its treatment

19. **Risk definition:** a measure of the expected losses due to a hazard event occurring in a given area over a specific time period.
20. **Working relation:** Risk = Probability of Hazard x Degree of Vulnerability - a heuristic that organises reasoning, never a validated predictive formula.
21. **Four treatment options:** risk acceptance (informed decision to bear the consequence), risk avoidance (do not undertake the risk-generating activity), risk reduction (lower the likelihood or consequence), risk transfer (shift the burden to another party, for example insurance).
22. **Boundary:** only avoidance and reduction change the loss that occurs; transfer changes who pays. Residual risk always remains.

The management cycle and the 3Rs to 3Ps shift

23. **Risk Management Phase (pre-disaster):** prevention, mitigation, preparedness.
24. **Crisis Management Phase (post-disaster):** relief, response, rehabilitation, reconstruction, recovery.
25. **3 Rs (traditional, reactive):** Rescue, Relief, Restoration.
26. **3 Ps (replacement, anticipatory):** Prevention, Preparedness, Proofing.
27. **Why it matters:** "restoration" reinstates the condition that failed; "proofing" writes risk reduction into recovery itself.

Pre-impact and post-impact instruments

28. **Prevention:** avoid the risk where feasible, for example by prohibiting construction in an active floodway or relocating a settlement.
29. **Mitigation, structural:** embankments, retrofitting, cyclone shelters.
30. **Mitigation, non-structural:** land-use zoning, building codes, insurance conditions, awareness.
31. **Preparedness:** contingency plans, stockpiles, early-warning systems, drills, trained response teams.
32. **Post-impact ladder:** relief and response (life safety), rehabilitation (services and livelihoods), reconstruction (physical assets), recovery (process of restoring or improving livelihoods, health, assets and systems).
33. **Build Back Better:** reconstruction must reduce future risk - safer siting, stronger standards, restored ecosystem buffers, protected livelihoods - rather than reproduce prior vulnerability.

34. **Risk-creation loop:** development choices raise exposure and vulnerability even where hazard frequency is unchanged; the recovery choice then either recreates or reduces that risk for the next cycle.

Disaster risk reduction

35. **Definition:** reducing disaster risks through reduced exposure to hazards, lessened vulnerability of people and property, wise management of land and the environment, and improved preparedness.
36. **Scope:** prevent new risk, reduce existing risk, manage residual risk.
37. **Field of action, in sequence with feedback:** policy and legal framework, risk assessment, risk awareness, plan implementation, early warning systems, use of knowledge.
38. **Diagnostic use:** identify the broken stage rather than reciting all six; downstream stages consume the outputs of risk assessment.

Resilience

39. **Definition verbs:** resist, absorb, accommodate, adapt, transform, recover - in a timely and efficient manner, preserving and restoring essential basic structures and functions.
40. **Determination:** the hazard-exposure-vulnerability-capacity configuration; lower vulnerability and higher capacity raise resilience for a given hazard and exposure.
41. **Framework elements:** risk knowledge, robustness, redundancy, resourcefulness, rapid recovery, adaptive capacity.
42. **Measurement limit:** resilience requires data on capacity, which is not operationalised with named indicators in the source vocabulary; state the metric or withhold the claim.

India's instruments and the National Disaster Management Plan

43. **Disaster Management Act, 2005:** statutory architecture of authorities, plans and funds - pre-Sendai.
44. **National Policy on Disaster Management, 2009:** consolidation of the prevention, mitigation and preparedness orientation - pre-Sendai.
45. **National Disaster Management Plan, 2016:** first edition; carries the four Sendai priorities into five Thematic Areas for Action - Understanding Risk, Inter-Agency Coordination, Investing in DRR (structural), Investing in DRR (non-structural), Capacity Development.
46. **Recorded critique of the 2016 edition:** too generic, no precise time frame, no projected fund requirement and, unlike Sendai or the Sustainable Development Goals, no goals or targets of its own.
47. **National Disaster Management Plan, 2019:** the revised and latest published edition; cite it as operative and do not carry the 2016 critique forward without a dated comparison.
48. **Prime Minister's ten-point agenda on disaster risk reduction:** announced at the Asian Ministerial Conference on Disaster Risk Reduction, New Delhi, November 2016, with each point aligned to one or more Sendai priorities of action.

From Hyogo to Sendai

49. **Hyogo Framework for Action 2005-2015:** five priorities, no quantified global targets, voluntary.
50. **Sendai Framework for Disaster Risk Reduction 2015-2030:** adopted at the Third UN World Conference on Disaster Risk Reduction, Sendai, Japan, on 18 March 2015; voluntary and non-binding; described as the first major agreement of the post-2015 development agenda.
51. **Four priorities for action:** understanding disaster risk; strengthening disaster risk governance to manage disaster risk; investing in disaster risk reduction for resilience; enhancing disaster preparedness for effective response and to Build Back Better in recovery, rehabilitation and reconstruction.
52. **Differences from Hyogo:** five priorities become four; seven measurable global targets are added; scope widens to man-made, technological and biological hazards; emphasis shifts from managing disasters to managing disaster risk; legal character is unchanged.
53. **Chronology discipline:** Sendai (18 March 2015) preceded the 2030 Agenda (September 2015) and the Paris Agreement (December 2015), so its text could not recognise them; coherence across the three is stressed in later

UNDRR material, including the 2023 Midterm Review.

The seven global targets and current anchors

54. **Targets A-D, loss outcomes:** A mortality; B affected people; C direct economic loss relative to global GDP; D damage to critical infrastructure and disruption of basic services including health and education.
55. **Measurement note:** Targets A and B are measured as per-100,000 decadal averages for 2020-2030 against the 2005-2015 baseline.
56. **Targets E-G, means of implementation:** E national and local DRR strategies by 2020; F international cooperation to developing countries; G availability of and access to multi-hazard early warning systems and disaster risk information and assessments.
57. **Deadline rule:** Target E alone carries a 2020 deadline; all others are 2030.
58. **Asymmetry:** A-D measure outcomes no state controls unilaterally; E-G measure means of implementation, but only some inputs such as strategy adoption are directly attributable to a State, while cooperation and warning access depend on multiple actors.
59. **Early Warnings for All:** UN Secretary-General's call of March 2022, launched at COP27 in November 2022, universal early-warning coverage targeted by 2027; pillars led by UNDRR (risk knowledge), WMO (observation and forecasting), ITU (dissemination), IFRC (preparedness and response), with WMO and UNDRR co-leading.
60. **Midterm Review:** High-Level Meeting held 17-19 May 2023; its Political Declaration was issued as UN General Assembly resolution 77/289 on 18 May 2023 - the correct dated anchor for any global-progress claim.

Examiner firewalls

- Do not equate a hazard with a disaster, or exposure with vulnerability.
- Do not present the risk relation as a precise universal formula or as a forecast.
- Do not merge prevention, mitigation and preparedness, or relief, rehabilitation and reconstruction.
- Do not describe insurance as risk reduction.
- Do not define resilience as bouncing back, or omit adaptation and transformation.
- Do not describe Sendai as a binding treaty or attach a 2020 deadline to any target except E.
- Do not confuse Sendai's four priorities with its seven targets, or global targets with national indicators.
- Do not treat EW4All or a national strategy count as proof of achieved coverage or reduced losses.
- Do not cite the 2016 National Disaster Management Plan as the current plan.
- Do not claim current target progress without the 2023 Midterm Review or another dated official source.

Answer spine for this topic

```
DEFINE HAZARD, EXPOSURE, VULNERABILITY, CAPACITY, RISK AND DISASTER
   -> USE THE RISK RELATION ONLY AS A HEURISTIC, WITH AREA AND PERIOD
   -> LOCATE THE PHASE: RISK MANAGEMENT OR CRISIS MANAGEMENT
   -> NAME THE INSTRUMENT: AVOIDANCE, REDUCTION, TRANSFER, ACCEPTANCE
   -> DEFINE RESILIENCE WITH SIX VERBS AND SIX FRAMEWORK ELEMENTS
   -> DATE THE INDIAN INSTRUMENTS BEFORE AND AFTER SENDAI
   -> COMPARE HYOGO AND SENDAI ON COUNT, TARGETS, SCOPE, LEGAL FORCE
   -> NAME TARGETS A-G WITH THE A-D / E-G SPLIT AND TARGET E'S 2020 DATE
   -> CLOSE AT THE LAST VERIFIED RUNG: INPUTS ARE NOT OUTCOMES
```

Evidence boundary

The adopted Sendai text is the anchor for the framework's architecture; the 2023 Midterm Review Political Declaration, issued as UN General Assembly resolution 77/289, is the anchor for implementation status; and Early Warnings for All is the anchor for how Target G is being pursued. None of these establishes avoided losses, country performance or universal coverage, and no such claim should be made without a separate dated source.

COMPLETE TOPIC ASCII MASTER FLOW DIAGRAM

ASCII MASTER FLOW - PANEL 1/12: Risk grammar and the disaster threshold

```
HAZARD -> dangerous condition or event with potential to harm
EXPOSURE -> people, livelihoods and assets in its path
VULNERABILITY -> conditions that turn exposure into harm
CAPACITY -> strengths, resources, institutions and measures
INTERACTION -> DISASTER RISK
THRESHOLD -> losses "beyond a community's ability to cope" = DISASTER
RULE -> same hazard magnitude, different outcome, because of E, V and C
```

ASCII MASTER FLOW - PANEL 2/12: Hazard taxonomy by source

```
NATURAL
  GEOPHYSICAL -> earthquake, tsunami, volcanic activity, landslide
  HYDROLOGICAL -> flood, flash flood, debris flow
  METEOROLOGICAL -> cyclone, storm, extreme temperature
  CLIMATOLOGICAL -> drought, desertification, forest fire
  BIOLOGICAL -> epidemic, pest infestation
SOCIO-NATURAL -> natural process amplified by human decisions
ANTHROPOGENIC -> pollution, deforestation, chemical spillage, accidents
USE -> family choice fixes the dominant mitigation instrument
```

ASCII MASTER FLOW - PANEL 3/12: Onset speed and creeping emergencies

```
RAPID ONSET -> earthquake, cyclone, flood, tsunami
  definable impact moment -> warning, evacuation, rescue, relief
SLOW ONSET / CREEPING EMERGENCIES -> climate change, desertification,
  soil degradation, drought
  no impact moment -> progressive loss of the ability to derive sustenance
  requirement -> must find adequate reflection in preparedness planning
GAP -> institutions built for acute events under-serve the slow category
```

ASCII MASTER FLOW - PANEL 4/12: The five-fold vulnerability typology

```
ECONOMIC -> income diversity, access to means of production
PHYSICAL -> proximity to hazard, construction type
SOCIAL -> poor, pregnant and lactating women, disabled, children, elderly
ENVIRONMENTAL -> ecosystem sensitivity, lost natural buffers
ATTITUDINAL -> negative disposition to change, dependence on outside help
RULE -> coinciding dimensions compound harm, they do not merely add
MARK -> the attitudinal dimension is the one most answers omit
```

ASCII MASTER FLOW - PANEL 5/12: Risk definition and the four treatment options

```
RISK = expected losses | GIVEN AREA | SPECIFIC TIME PERIOD
WORKING RELATION = PROBABILITY OF HAZARD x DEGREE OF VULNERABILITY
USE -> order the problem and locate the movable factor
DO NOT USE -> arithmetic prediction of the next event
TREATMENT
  ACCEPTANCE -> informed decision to bear the consequence
  AVOIDANCE -> do not undertake the risk-generating activity
  REDUCTION -> lower the likelihood or the consequence
  TRANSFER -> shift the burden, for example insurance
LIMIT -> only avoidance and reduction change the loss; residual risk remains
```

ASCII MASTER FLOW - PANEL 6/12: Two-phase cycle and the 3Rs to 3Ps shift

```
PRE-DISASTER: RISK MANAGEMENT PHASE
  PREVENTION -> MITIGATION -> PREPAREDNESS
IMPACT
POST-DISASTER: CRISIS MANAGEMENT PHASE
  RELIEF -> RESPONSE -> REHABILITATION -> RECONSTRUCTION -> RECOVERY
APPROACH SHIFT
  OLD 3 Rs -> RESCUE, RELIEF, RESTORATION (reactive)
  NEW 3 Ps -> PREVENTION, PREPAREDNESS, PROOFING (anticipatory)
POINT -> restoration reinstates the failed condition; proofing does not
```

ASCII MASTER FLOW - PANEL 7/12: Pre-impact instruments

```
PREVENTION -> can the risk be avoided? relocation, prohibition in floodway
MITIGATION -> can the impact be lessened?
  STRUCTURAL -> embankments, retrofitting, cyclone shelters
  NON-STRUCTURAL -> land-use zoning, building codes, awareness, incentives
PREPAREDNESS -> can we act in time? plans, stocks, warning, drills, teams
SEQUENCE -> the three are complementary, never substitutable
TRAP -> relocation is prevention, not mitigation
```

ASCII MASTER FLOW - PANEL 8/12: Post-impact ladder and the recovery fork

```
RELIEF AND RESPONSE -> life safety, medical care, shelter, water
REHABILITATION -> services, schooling, health access, livelihoods
RECONSTRUCTION -> housing, infrastructure, productive assets
RECOVERY -> process of restoring or improving functioning; sets the next risk profile
FORK
  RESTORE AS BEFORE -> the same vulnerability returns
  BUILD BACK BETTER -> safer siting, stronger codes, restored buffers
RISK-CREATION LOOP -> development choice raises exposure and vulnerability
  even where hazard frequency is unchanged; recovery recreates or reduces it
```

ASCII MASTER FLOW - PANEL 9/12: DRR definition and field of action

```
DRR = reduced EXPOSURE + lessened VULNERABILITY of people and property
      + wise management of LAND and ENVIRONMENT + improved PREPAREDNESS
SCOPE -> prevent new risk, reduce existing risk, manage residual risk
FIELD OF ACTION (sequence with feedback)
  POLICY AND LEGAL FRAMEWORK -> RISK ASSESSMENT -> RISK AWARENESS
  -> PLAN IMPLEMENTATION -> EARLY WARNING SYSTEMS -> USE OF KNOWLEDGE
DIAGNOSTIC -> name the broken stage; everything downstream consumes
  the output of risk assessment
```

ASCII MASTER FLOW - PANEL 10/12: Resilience, elements and measurement limit

```
RESILIENCE VERBS -> RESIST, ABSORB, ACCOMMODATE, ADAPT, TRANSFORM, RECOVER
  in a timely and efficient manner, preserving and restoring essential
  basic structures and functions
DETERMINED BY -> hazard, exposure, vulnerability and capacity configuration
FRAMEWORK ELEMENTS -> RISK KNOWLEDGE, ROBUSTNESS, REDUNDANCY,
  RESOURCEFULNESS, RAPID RECOVERY, ADAPTIVE CAPACITY
TRAP -> "bouncing back" restores the condition that failed
LIMIT -> measurement needs capacity data and a stated metric
```

ASCII MASTER FLOW - PANEL 11/12: India's instruments and the NDMP

```
BEFORE SENDAI
  DISASTER MANAGEMENT ACT, 2005 -> statutory architecture
  NATIONAL POLICY ON DISASTER MANAGEMENT, 2009 -> prevention orientation
AFTER SENDAI
  NDMP 2016 -> five Thematic Areas for Action
    UNDERSTANDING RISK | INTER-AGENCY COORDINATION
    INVESTING IN DRR (STRUCTURAL) | INVESTING IN DRR (NON-STRUCTURAL)
    CAPACITY DEVELOPMENT
  PM TEN-POINT AGENDA -> AMCDRR, NEW DELHI, NOVEMBER 2016,
    each point aligned to Sendai priorities of action
  NDMP 2019 -> revised and latest published edition; cite as operative
CRITIQUE OF 2016 EDITION -> too generic, no time frame, no fund projection,
  no goals or targets of its own
```

ASCII MASTER FLOW - PANEL 12/12: Hyogo to Sendai, Targets A-G and the answer boundary

```
HYOGO 2005-2015 -> five priorities, no quantified targets, voluntary
SENDAI 2015-2030 -> adopted 18 MARCH 2015, Third UN World Conference,
  Sendai, Japan; voluntary and non-binding; four priorities
  1 UNDERSTAND RISK | 2 STRENGTHEN RISK GOVERNANCE
  3 INVEST IN DRR FOR RESILIENCE | 4 PREPARE AND BUILD BACK BETTER
TARGETS
  A mortality | B affected people | C economic loss / global GDP
  D critical infrastructure and basic services      -> loss outcomes, 2030
  E national and local DRR strategies by 2020       -> only 2020 deadline
  F international cooperation | G multi-hazard early warning and risk
  information                                       -> means of implementation
CURRENT ANCHORS -> EW4All (call March 2022, COP27 November 2022, by 2027;
  UNDRR, WMO, ITU, IFRC pillars) | MIDTERM REVIEW 17-19 MAY 2023,
  Political Declaration as UNGA RESOLUTION 77/289, 18 MAY 2023
BOUNDARY -> frameworks, strategies and initiatives are inputs;
  avoided loss and coverage require separate dated outcome evidence
```